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(ICS 97.220.10; 97.220.40)

SINGAPORE STANDARD

Code of practice for outdoor adventure education (OAE) activities

Incorporating Corrigendum No. 1



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Foreword

This Singapore Standard was prepared by the Working Group on Outdoor Adventure Education set up by the Technical Committee on Personal Safety and Health under the purview of the Safety and Quality Standards Committee. It incorporates Corrigendum No. 1, August 2025 denoted by  .

This standard specifies minimum requirements and best practices for safe and responsible planning and delivery of OAE activities, including the management of risk and the prevention of injuries and fatalities.

This standard helps providers of OAE programmes to develop appropriately managed educational adventure activities for individuals and the communities, while protecting the environment and culturally significant places.

It is presupposed that in the course of their work, users will comply with all relevant regulatory and statutory requirements. Some examples of relevant regulations and acts are listed in the Bibliography. The Singapore Standards Council and Enterprise Singapore shall not be responsible for identifying all of such legal obligations.

In preparing this standard, reference was made to the following publications:

1. ANSI/ACCT 03-2019 Challenge courses and canopy/zip line tours standards
2. SS 570-1:2022 Personal protective equipment for protection against falls from height – Part 1: Single-point anchor devices
3. SS 570-2:2022 Personal protective equipment for protection against falls from height – Part 2: Flexible horizontal lifeline systems
4. SS 607:2015 Specification for design of active fall-protection systems
5. SS 681:2022 Code of practice for sport safety
6. SS 701:2023 Code of practice for inland and open water sporting activities
7. Adventure based learning (ABL) activities, Good practice guide, Version 2, 2021, Education Outdoors New Zealand, Recreation Aotearoa Te Whai Oranga
8. Best practices on emergency action plan (EAP) for outdoor adventure education (OAE) programmes & operations, 2023, Outdoor Adventure Education (OAE) Council

Acknowledgement is made for the use of information from the above publications.

Attention is drawn to the possibility that some of the elements of this Singapore Standard may be the subject of patent rights. Enterprise Singapore shall not be held responsible for identifying any or all such patent rights.

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NOTE

1. *Singapore Standards (SSs) and Technical References (TRs) are reviewed periodically to keep abreast of technical changes, technological developments and industry practices. The changes are documented through the issue of either amendments or revisions. Where SSs are deemed to be stable, i.e. no foreseeable changes in them, they will be classified as “mature standards”. Mature standards will not be subject to further review unless there are requests to review such standards.*
2. *An SS or TR is voluntary in nature except when it is made mandatory by a regulatory authority. It can also be cited in contracts making its application a business necessity. Users are advised to assess and determine whether the SS or TR is suitable for their intended use or purpose. If required, they should refer to the relevant professionals or experts for advice on the use of the document. Enterprise Singapore and the Singapore Standards Council shall not be liable for any damages whether directly or indirectly suffered by anyone or any organisation as a result of the use of any SS or TR. Although care has been taken to draft this standard, users are also advised to ensure that they apply the information after due diligence.*
3. *Compliance with a SS or TR does not exempt users from any legal obligations.*

Code of practice for outdoor adventure education (OAE) activities

0 Introduction

OAE activities provide a platform for education in a practical environment. Skills such as critical thinking, leadership, teamwork and problem solving can all be taught and applied practically, adding depth to the education experience. This standard provides activity providers with requirements and recommendations for planning and managing these activities so that they can be conducted in a safe and effective way.

The following trends in OAE activities highlight the need for a structured approach to their planning:

- a) An increasing interest in providing education in engaging and effective ways.
- b) A strong emphasis on safety.

1 Scope

This standard is designed to help OAE activity providers deliver safe and educational experiences. It establishes a framework and describes best practices which can be used for creating or revising OAE activity programmes while highlighting the unique safety requirements for the most common outdoor adventure activity types used for educational purposes in Singapore.

This standard covers the following:

- a) Accountabilities of OAE activity providers;
- b) Responsibilities and minimum competence of programme supervisors, activity leaders and assistant activity leaders;
- c) Risk management system;
- d) Readiness of locations and facilities; and
- e) Adequacies of equipment and the minimum information to be communicated to participants and potential participants before, during and after the activity for their safety.

While the information provided in this standard may be useful in planning and managing other activities, it only covers the OAE activities explicitly mentioned.

2 Normative references

The following referenced documents are indispensable for the application of this standard. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ANSI/ACCT 03	Challenge courses and canopy/zip line tours standards
ANSI Z359	Fall protection and fall restraint
ASTM 1772	Standard specification for harnesses for rescue and sport activities
EN 12277	Mountaineering harnesses

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EN 12572-1	Artificial climbing structures – Part 1: Safety requirements and test methods for ACS with protection points
EN 12572-2	Artificial climbing structures – Part 2: Safety requirements and test methods for bouldering walls
EN 12572-3	Artificial climbing structures – Part 3: Safety requirements and test methods for climbing holds
EN 15567-1	Sports and recreational facilities – Ropes courses – Part 1: Construction and safety requirements
EN 15567-2	Sports- and recreational facilities – Ropes courses – Part 2: Operation requirements
ISO 12402-5	Personal flotation devices – Part 5: Buoyancy aids (level 50) – Safety requirements
NFPA 1983	Standard on life safety rope and equipment for emergency services
UIAA 105	Harnesses recommendations for inspection and retirement

3 Terms and definitions

For the purpose of this Singapore Standard, the following terms and definitions apply.

3.1 activity

a structured task or series of related tasks for OAE

3.2 activity leader

a competent person leading and supervising an activity

3.3 activity leadership

refers to the activity leaders and assistant activity leaders involved in an activity

3.4 activity provider / provider

any individual or organisation including businesses, community groups, government agencies, schools or any other group, that implements OAE activities with participants

3.5 assistant activity leader

a competent person who helps lead and supervise an activity, under the direction of an activity leader

3.6 client organisation

the client of an activity provider is someone other than the participants undertaking the activity

Note 1 to entry: For example, a school uses a provider to deliver an activity for its students, in this case, the school is the client organisation and the students are participants.

3.7 competent person

a person who has the required skill level, qualifications and experience for the duties to which they are assigned

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3.8 direct supervision

where those responsible for supervising participants during all or part of the adventure activity are in close proximity to the participants and is able to intervene immediately

3.9 emergency position indicating radio beacon (EPIRB)

a tracking transmitter which aids in the detection and location of boats, aircraft and people in distress

3.10 facilitation

enabling a clear and easy-to-understand link between a participant's experience in an OAE programme and its activities, and the intended learning outcomes

3.11 incident

an occurrence which has caused, or had the potential to cause, injury to people or damage to property

3.12 indirect supervision

where those responsible for supervising participants during all or part of an adventure activity are in the general vicinity of the participants but are not able to intervene immediately

3.13 inland water

a body of water completely enclosed by land

Note 1 to entry: This includes reservoirs, rivers and quarry lakes.

3.14 land manager

the party that is responsible for managing a property, usually either the owner or a party appointed by the owner

Note 1 to entry: This can include the power to restrict access, place conditions of entry or requirements on anyone accessing that property.

3.15 mass capsized event

when multiple craft capsize within a short period of time, resulting in a large number of participants in the water

3.16 non-participating contact

a person who is nominated to act on behalf of those undertaking the activity in accordance with the emergency management plan

Note 1 to entry: This person is neither directly involved in the activity nor located with those conducting it.

3.17 non-remote location

any location that is not a remote location (see 3.24)

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3.18 open water

a large body of water

Note 1 to entry: This includes channels, fairways, beach fronts, coastal areas, lagoons, bays and wetlands.

3.19 participant

a person who is reliant upon the activity leaders and assistant activity leaders for supervision, guidance or instruction to support that person's participation in an OAE activity

3.20 personal locator beacon (PLB)

a transmitter that aids in the detection and location of boats, aircraft and people in distress

3.21 pit toilet

a temporary toilet constructed by digging a pit in the earth and is filled in and covered over after use

3.22 programme supervisor

a competent person who is in charge of the programme

Note 1 to entry: Their responsibilities include managing the programme schedule plan, supervising the activity leadership, duty personnel and activity changes, providing for the participants' basic needs, and incident and emergency management.

3.23 qualified peer

a person who is sufficiently experienced to assess the competence of prospective activity leaders and assistant activity leaders

Note 1 to entry: The qualified peer is either certified in the competency being assessed or is recognised by the industry as an experienced and competent person in the competency being assessed where no certification exists.

3.24 remote location

a place that is more than 30 mins from access to reliable telephone access, 20 mins from vehicle access or 1 h from access to professional medical assistance in all foreseeable weather

3.25 remote supervision

where those responsible for supervising participants during all or part of the adventure activity, are not involved in direct or indirect supervision, unlikely to be in the vicinity, and although immediately contactable can take time to respond in person

3.26 responsible person

a competent person who is assigned tasks or duties during an activity that do not require the activity-specific competence of an activity leader or assistant activity leader

3.27 spotting

a learned and practiced technique involving one person or multiple persons working together to monitor a person and intervene physically when required

Note 1 to entry: Physical intervention includes providing physical support, lifting, cushioning, and catching the participant, or otherwise minimising the effects of a fall.

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3.28 vulnerable participant

a participant who is a minor or an individual who is unable to take care of themselves, or is unable to protect themselves against harm or exploitation by reason of age, illness, trauma, disability or any other reason

3.29 use of “shall”, “should”, “may”, and “can”

In this standard, the following verbal forms are used:

- “shall” indicates that the requirement is strictly to be followed in order to conform to the standard and from which no deviation is permitted;
- “should” indicates a recommendation;
- “may” indicates a permission;
- “can” indicates a possibility or a capacity.

4 Core planning principles

4.1 Programme planning

4.1.1 About

An OAE programme consists of one or more OAE activities. Planning is done at both programme and activity level. Figure 1 illustrates content that is typically included in activity-level and programme-level plans.

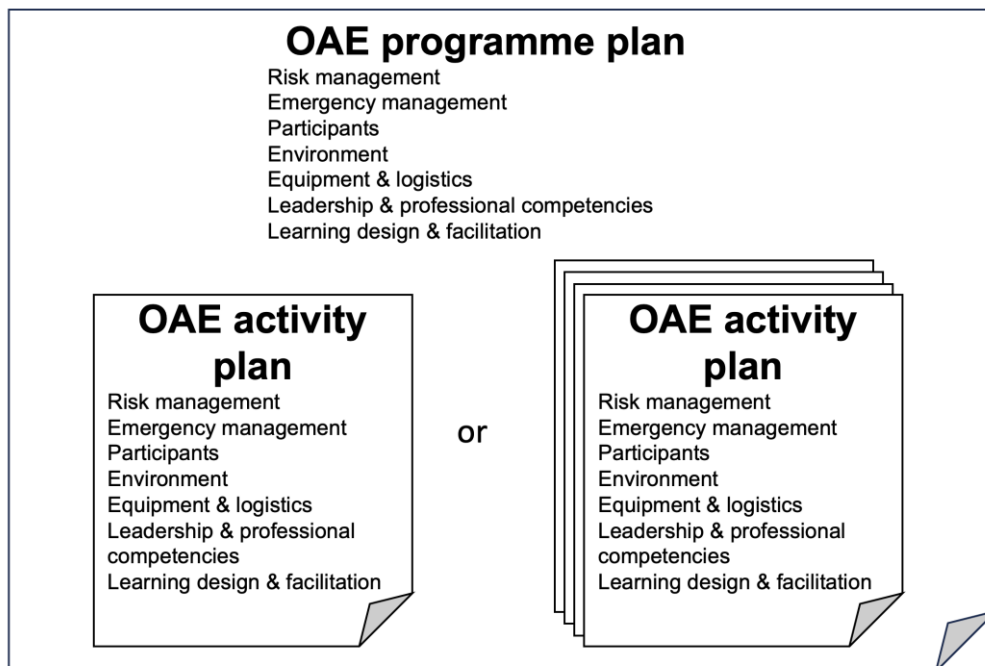


Figure 1 – OAE programme planning overview

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4.1.2 Components of a programme plan

4.1.2.1 Programme planning shall include but is not limited to the following:

- a) Determining the educational aims of the programme;
- b) Learning design (see 4.9);
- c) Gathering relevant information on the intended participants (see 4.5.3);
- d) Determining which activities are part of the programme;
- e) Scheduling the activities;
- f) Risk management plans (see 4.3);
- g) Emergency management plans (see 4.4); and
- h) An activity plan for each included activity (see 4.2).

4.1.2.2 Programme planning should include but is not limited to the following:

- a) Coordination of participants' arrival to the programme, transportation between activities and safe return when the programme has finished;
- b) Formal code of ethics; and
- c) All other programme elements which are not included in the activity plans.

4.1.2.3 The activities to include in an OAE programme shall be determined by the provider based on:

- a) client organisation's requirements;
- b) suitability of the activity for the participants;
- c) effectiveness of the activity to achieve one or more identified educational aims; and
- d) availability of the activity resources.

4.1.2.4 Further information on planning programmes and activities can be found in 4.2 to 4.9, including guidelines and requirements applicable to all activities detailed in Clauses 5 to 11.

4.1.3 Documentation

The provider shall keep all programme plans including associated activity plans for a minimum period of 3 years.

4.1.4 Process for approval

There shall be a formal, documented process for approving programme plans within the provider's organisation.

This process should include approval of the following documents:

- a) Programme plan which comprises:
 - a risk management plan;
 - an emergency management plan;
 - details of programme coordinators; and
 - a programme learning design and facilitation plan.
- b) Activity plan (see 4.2) for each included activity.

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4.2 Activity planning

4.2.1 Objectives

The provider shall create an activity plan for each activity in their OAE programme and include it in the programme plan.

The activity plan shall define the purpose and objectives of an activity and help to inform decisions made by the activity leadership and participants before, during and after the activity. A process for developing activity plans that adapts to the individual context, and the level of details required for each activity shall be implemented.

All activity plans shall be:

- a) approved before implementation (see 4.1.4); and
- b) documented.

4.2.2 Components of an activity plan

At a minimum, the provider shall include the following in each OAE activity plan:

- a) Activity schedule;
- b) Risk management plan (see 4.3);
- c) Emergency management plan (see 4.4);
- d) Details of the participants (see 4.5);
- e) Environmental considerations (see 4.6);
- f) Equipment and logistics plan (see 4.7);
- g) Leadership and professional competencies (see 4.8); and
- h) Learning design and facilitation plan (see 4.9).

In addition, a contingency plan (see 4.3) should also be developed for each activity to take into account unforeseen disruptions.

4.2.3 Activity plan framework

The activity plan shall be based broadly on the type and intent of the activity, and then expanded to include activity-specific details for each of the areas as shown in Figure 2.

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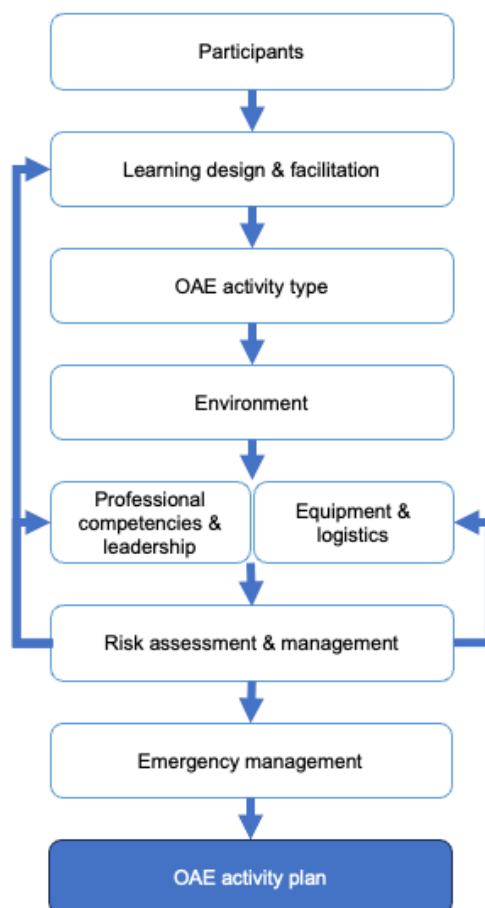


Figure 2 – Activity planning overview

4.2.4 Documentation and information sharing

4.2.4.1 To facilitate implementation of the activity, the provider shall ensure timely access to all relevant documentation for the activity leadership and, where applicable, external non-participating contacts (e.g. teachers-in-charge) to assist them prepare, deliver and manage the activity.

4.2.4.2 This documentation shall include, but is not limited to the following:

- a) Activity plan;
- b) Communications plan;
- c) Operating procedures;
- d) All relevant supporting documents including:
 - charts;
 - maps; and
 - checklists;
- e) Current medical information for all participants under the activity leader's charge (see 4.5.8).

4.2.4.3 Other information that should be made available to the activity leadership includes:

- a) the participants' consent to medical treatment (see 4.5.4); and
- b) consent for vulnerable persons to participate (see 4.5.7);

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4.2.4.4 The provider should only furnish the following information when specifically requested by the relevant authorities:

- a) Names and emergency contact details for:
 - participants;
 - activity leader and assistant activity leaders; and
 - responsible persons; and
- b) Physical copies of licences, approvals and access permits.

4.2.4.5 The provider should only furnish the following information when specifically requested by the client organisation:

- a) Names and emergency contact details for:
 - activity leader and assistant activity leaders; and
 - responsible persons; and
- b) Physical copies of licences, approvals and access permits.

4.3 Risk management plan

4.3.1 Establishing a risk management plan

Risk management is a systematic approach to identifying and mitigating or eliminating risks and is an integral component of a well-planned and safe OAE programme.

Risk management shall be integrated into all aspects of providers' governance, management and operations. This ensures that it is embedded into the policies and practices of the provider, and it covers all factors that contribute to risks, not just those occurring during the delivery of the activity itself.

The provider shall include a documented risk management plan in all their OAE programmes. The risk management plan shall consist of:

- a) an overall risk management plan for the programme; and
- b) individual risk management plans for each activity within the programme.

4.3.2 Scope of the risk management plan

The risk management process shall factor in all parties directly or indirectly involved in the OAE programme. Insufficient or incomplete information from any of these parties can negatively impact the effectiveness of the risk management plan. Directly and indirectly involved parties include, but are not limited to the following:

- a) Provider;
- b) Activity leadership;
- c) Responsible persons;
- d) Participants;
- e) Client organisations; and
- f) Parents and legal guardians.

A risk management framework (see 4.3.3) should be used to provide a structure to guide effective risk management.

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4.3.3 Risk management framework

The risk management framework includes the identification of risks, establishment of mitigating measures after analysis and evaluation of risks, communication of mitigating measures to necessary parties, and a continuous review of the risk assessment and management using documentation created at each stage. Figure 3 illustrates the broad framework for risk management.

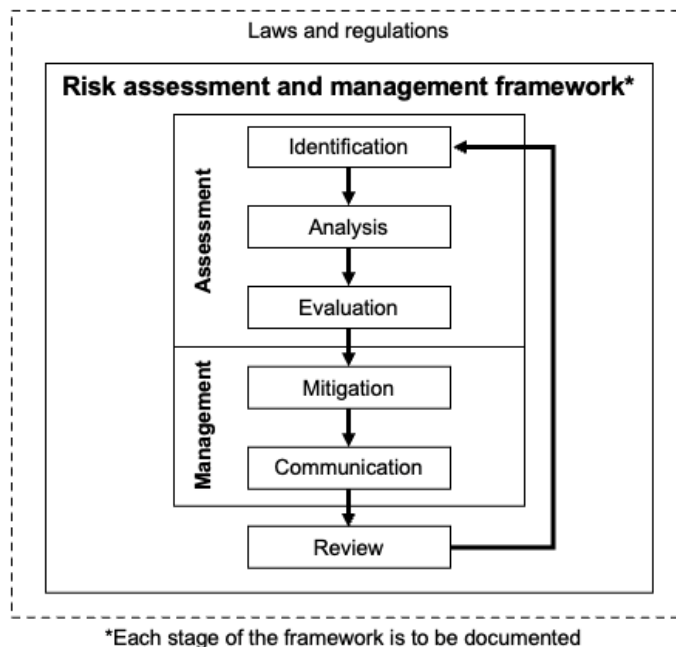


Figure 3 – Risk assessment and management framework for OAE programmes

The steps for formulating the risk management plan shall include the following:

- Identifying hazards systematically;
- Analysing identified hazards to map out associated risks;
- Identifying and evaluating risks not associated with identified hazards;
- Categorising risks (to facilitate risk mitigation);
- Developing procedures to mitigate risks;
- Communicating risk mitigation procedures;
- Implementing risk mitigation procedures; and
- Reviewing the risk management plan.

4.3.4 Risk management process

In assessing and managing risks, the potential harm associated with the OAE programme shall be evaluated by the provider.

The risk management process should include:

- referring to past incidents;
- adhering to best practices and standards;
- consulting experts where necessary; and
- consulting stakeholders.

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The steps of the risk management process are described in Table 1.

Table 1 – Risk management steps

Step	Description
Risk identification	The potential hazards associated with the OAE programme shall be systematically scanned.
Risk analysis	An estimate of the likelihood and severity of the harm to individuals and environments shall be carried out.
Risk evaluation	Agreement with client organisations on accepting or mitigating the identified risks.
Development of risk mitigation	Control measures to reduce or eliminate risks to levels that are mutually acceptable to client organisations and provider shall be developed.
Implementation of risk mitigation	Implementation of control measures to reduce or eliminate the risks, including the deployment of resources, roles and responsibilities.
Monitoring risks	Monitoring the effectiveness of risk controls and adjusting them when necessary.
Communication of risks	Communicating information and status of risks before and during an OAE programme to the client organisation, parents, participants, and other associated parties.
Documentation of risk assessment and management	Documentation and retention of the risk assessment and management processes.
Continuous review of risk assessment and management	Reviewing the process and documentation for the risk assessment and management: a) every 3 year(s); b) in the event of an incident that requires an update; or when there is a change in the OAE programme scope.

Annex A gives descriptions of medical conditions which can affect a person's ability to participate in an activity and should be factored into the risk management plan where relevant.

4.3.5 Communication of the risk management strategy

The risk management strategy can be communicated through:

- a) company policies;
- b) activity standard operating procedures (SOPs); and
- c) other programme instructions.

The provider shall furnish the activity leadership with the full risk management strategy. The provider shall ensure the client organisation and participants are briefed on the components of the risk management strategy applicable to them, especially the part they play in it.

4.3.6 Contingency planning

As part of both programme and activity planning, preparations should be made for sudden disruptions to the implementation of the programme and activity plans, including, but not limited to:

- a) unfavourable environment conditions;

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- b) reductions in manpower; and
- c) changes to available resources.

The contingency plan can describe alternative locations, activities or resources and the criteria for triggering their use.

4.4 Emergency management

4.4.1 Emergencies in OAE

Emergencies can occur despite careful planning. Emergencies are often sudden, unexpected, can significantly affect individuals and groups, and require an immediate response.

Having a comprehensive emergency management plan (also known as an emergency action plan) in place facilitates a timely and effective response to emergencies.

Emergencies in the OAE context can include, but are not limited to incidents which result in or present a risk of:

- a) damage to equipment, property or the environment;
- b) any incident which may prevent the continuation of the activity;
- c) injury to people;
- d) injury to wildlife;
- e) illness; and
- f) death.

Such incidents can be the result of factors including, but not limited to:

- a) equipment failure;
- b) human error;
- c) unforeseen factors (e.g. accidental interference from a third party);
- d) adverse weather conditions; and
- e) natural disasters.

4.4.2 Key objectives in emergency management

Emergency management involves systematically:

- a) preparing for emergencies;
- b) responding to emergencies; and
- c) recovering from emergencies.

The aim of emergency management is to:

- a) eliminate or limit the severity of injuries, damage and loss arising from an incident;
- b) implement an immediate response to an incident to prevent further harm;
- c) prevent a situation from further deteriorating;
- d) resume normal operations at the earliest opportunity; and
- e) conduct post-incident recovery and subsequent resumption of affected activities.

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4.4.3 Emergency management planning levels

The provider shall include an emergency management plan in their OAE programmes, which shall consist of the following:

- a) An overall emergency management plan for the programme; and
- b) Individual emergency management plans for each activity within the programme.

Figure 4 illustrates the overview of an emergency management.

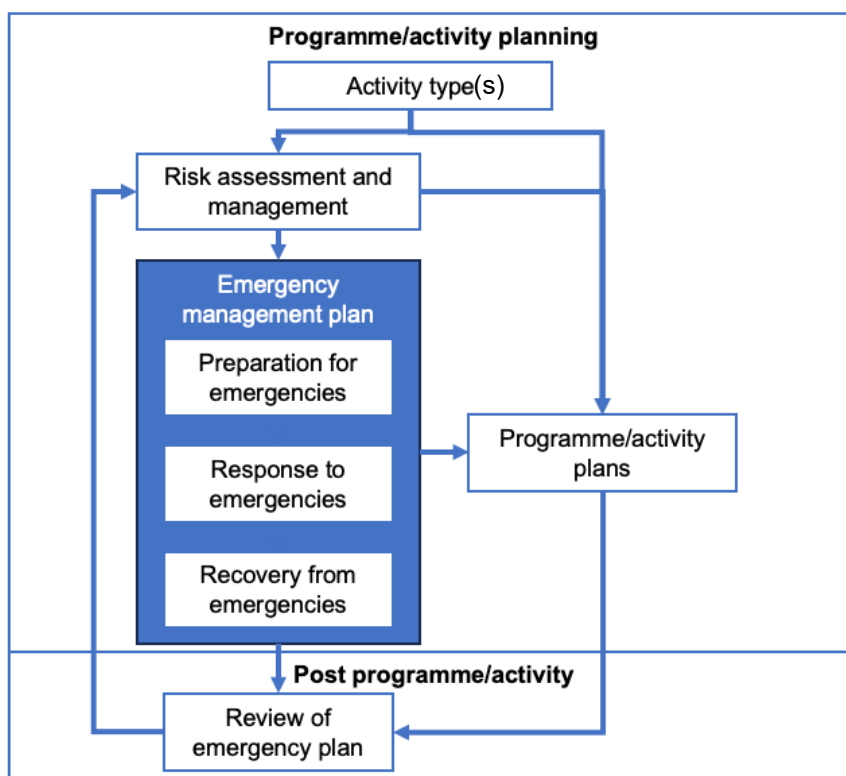


Figure 4 – Overview of emergency management

To effectively manage emergencies, the provider shall establish and include SOPs for emergencies in the emergency management plan. These SOPs shall include protocols for:

- a) providing first aid;
- a) evacuation from the incident location to medical services;
- b) internal communications to staff; and
- c) external communications to parties including:
 - client organisation;
 - relevant authorities;
 - media; and
 - other relevant stakeholders.

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4.4.4 Developing an emergency management plan

The process of developing the emergency management plan shall include:

- a) referencing to past incidents and best practices and standards, wherever applicable;
- b) consulting with the client organisation and aligning the emergency management plan to at least meet their requirements;
- c) consulting experts and stakeholders, wherever applicable; and
- d) documenting the entire process.

4.4.5 Scope of an emergency management plan

The emergency management plan established by the provider to prepare for foreseeable emergencies shall include, but not be limited to the following:

- a) Details of the types of emergencies which can foreseeably occur;
- b) Details on the actions to be taken for each type of emergency;
- c) Procedure for designating party/parties to be responsible for ensuring the identified actions are carried out correctly;
- d) Details on access to medical and emergency services and any other resources which can be foreseeably required;
- e) Evacuation plans covering all foreseeable emergencies and locations for the activity;
- f) Identification of:
 - emergency access points;
 - escape routes; and
 - assembly points;
- g) Criteria that determine the severity of foreseeable situations, and corresponding responses to the severity of the situations;
- h) Flowchart indicating priority actions; and
- i) Implementation procedure for emergency plans.

4.4.6 Effective use of an emergency management plan

An emergency management plan is most effective when all pertinent information has been included and all relevant parties are familiar with the plans. The activity leadership should run through the response plan before the start of each programme. To ensure that the emergency management plan can be used effectively, the provider shall:

- a) obtain information needed for risk assessment, risk management, and emergency management, including:
 - relevant medical histories, and
 - emergency contact numbers;
- b) communicate the emergency management plan to the client organisation;
- c) communicate relevant information to all personnel involved in the OAE programme on their duties and responsibilities during emergencies. This includes the:
 - activity leadership, and
 - participants;

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- d) ensure all responsible persons for an OAE programme are familiar with:
 - the emergency management plan, and
 - their roles and responsibilities in an emergency situation;
- e) test and evaluate the emergency management plan on a regular basis (see 4.4.7).

4.4.7 Emergency management plan reviews

The emergency management plan should be reviewed together with the client organisation and approved by them prior to implementation.

Emergency management plans and supporting documentation shall be reviewed by the provider after the initial deployment and at least once every three year(s) after that.

Gaps or weaknesses in emergency management plans can be detected either through the occurrence of an incident or a regular review process. Where such gaps are identified in either the emergency management plan or documentation, the provider shall address them:

- a) as soon as possible; and
- b) before the start of the next programme involving a similar activity.

4.4.8 Activating the emergency management plan

In the event of an emergency, the party/parties designated in the emergency management plan shall assume their respective emergency roles and undertake the responsibilities specified in the emergency management plan.

4.4.9 Incident reporting, recording and monitoring

Reporting of incidents and near misses helps identify risks and improve risk management for future iterations of the programme. The information accumulated from incident reports help to inform both the provider's organisation and the industry sector in general.

A clear and accurate record of all incidents that occur during an activity shall be recorded by the provider. This includes near misses, i.e. events or sets of circumstances that can foreseeably have led to an incident but did not.

The provider shall implement a process and set up a supporting system to keep records of incidents and near misses that are reported by the activity leadership and other employees. The following actions shall be taken by the provider as part of the implementation:

- a) Set up a system to objectively collect and record information and data for all incidents and near misses without analysis or rationalisation, including:
 - relevant medical information of any injured parties;
 - incident reports; and
 - after incident reports.
- b) Ensure that this system is correctly used by staff including sub-contractors (if any).
- c) Ensure that the data captured by the system is analysed:
 - by incident, immediately or as soon as practicable after the incident; and
 - collectively by categories of incident and activity on a periodic basis.
- d) Review the risk management plan for the activity in which the incident(s) occurred (see 4.3.4).

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- e) Revise the activity plans (see 4.2) where appropriate to reduce or eliminate the occurrence of similar incidents in future.
- f) Apply these revisions to any other activities which can incur a similar risk of such an incident occurring.

4.5 Participants

4.5.1 Types of participants

An OAE programme may include different types of participants who are engaged at varying levels based on their ability and willingness to partake in the different activities. Participants may range from those who can be assigned additional tasks or duties for an increased level of involvement to vulnerable participants who may not participate fully in all aspects of the programme.

Where a vulnerable participant is accompanied by a carer, the provider shall treat that carer as a participant for the duration of the programme. See 4.5.7 for more information on vulnerable participants.

4.5.2 Factors relating to participants

The needs and capabilities of all participants shall be factored into every stage of an OAE activity design, management, planning and delivery. This includes but is not limited to:

- a) taking all reasonable steps to understand the abilities and needs of individual participants, group and/or client organisation;
- b) designing an appropriate programme which accounts for the abilities and needs of all participants;
- c) implementing the programme and all included activities in a manner which accounts for the abilities and needs of all participants;
- d) providing, in a timely and effective manner, all necessary parties with:
 - all relevant information about the programme;
 - all relevant information about all activities in the programme; and
 - any participant requirements for each activity.

4.5.3 Pre-programme communication

4.5.3.1 Pre-programme communication is the two-way flow of information between the provider and potential participants and/or their legal guardians. It takes place prior to a participant agreeing to take part in a programme and includes information flow from the following:

- a) Provider to the participants/legal guardians and client organisation; and
- b) Participants/legal guardians and client organisation to the provider.

4.5.3.2 Information from the provider assists potential participants, parents/legal guardians and client organisations in making an informed decision regarding participation in an activity. This information shall include a comprehensive introduction to, and the requirements of, each activity within the planned OAE programme and shall be communicated to all participants prior to their agreeing to participate.

4.5.3.3 The introduction to the activities shall be:

- a) given according to the capacity of participants to comprehend, especially those with specific impairments e.g. cognitive, hearing, etc.;
- b) accurate at the time it is given;

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- c) relevant; and
- d) provided at an appropriate time.

4.5.3.4 The pre-programme communication from the provider can be delivered in one or multiple formats which can include, but are not limited to:

- printed documentation;
- live presentations;
- verbal discussions; and
- pre-recorded videos.

4.5.3.5 The information communicated pre-programme should include, but is not limited to the following:

- a) Objective(s), type, nature and duration of each included activity;
- b) Environments in which each included activity takes place;
- c) Participant responsibilities, including:
 - following directions;
 - appropriate behaviour; and
 - bringing any concerns regarding the safety, health and well-being of themselves and others to the attention of a member of the activity leadership;
- d) Personal attire (see 4.7.2);
- e) Any required personal equipment (see 4.7.1);
- f) Any equipment supplied by the provider (see 4.7);
- g) Physical capability required to participate safely;
- h) Provision of necessities including water, food and shelter;
- i) Pre-requisite skills (if any);
- j) Pre-requisite qualifications (if any);
- k) Any required or recommended prior experience;
- l) Foreseeable benefits associated with participation;
- m) Foreseeable risks associated with participation (see 4.3);
- n) Appropriate summary of the risk management strategies (see 4.3);
- o) Appropriate summary of the emergency response plan (see 4.4);
- p) Information regarding the provider's insurance and liability coverage; and
- q) Information on the provider, activity leaders and assistant leaders including:
 - their backgrounds;
 - their roles for the activity; and
 - relevant contact details.

4.5.3.6 The provider shall request information from the client organisation, intended participants or their legal guardians to enable the creation and implementation of a programme which is appropriate for the intended participants. Such information can include:

- a) a signed consent to receive medical treatment (see 4.5.4);

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- b) a signed acknowledgement of risk for each activity in the programme (see 4.5.5); and
- c) relevant details on pre-existing health conditions (see 4.5.8).

4.5.3.7 The participants and/or their legal guardians shall be informed by the provider of the potential consequences for:

- a) submitting incomplete information;
- b) submitting false or inaccurate information; and
- c) failing to submit the requested information.

4.5.3.8 Such consequences can include:

- a) the participant being at increased risk in an emergency situation;
- b) other participants being placed at risk; and
- c) the participant not being allowed to take part in activities.

4.5.4 Consent to medical treatment

Prior to attending a programme, the provider should obtain confirmation from all participants or their legal guardians that consent has been given to provide and seek medical treatment, if necessary.

Consent to medical treatment shall be:

- a) given in writing; and
- b) clearly communicated to the activity leadership for each activity in the programme.

Where consent to medical treatment has been given to the provider, the activity leadership who are representatives of the provider, may proceed with implementing the necessary medical treatment within the limits of the consent declaration.

Where consent to medical treatment has been given to the client organisation, a clear process shall be established by the provider, with the agreement of the client organisation, for addressing such medical situations.

4.5.5 Acknowledgement of risk

All physical activities come with some degree of risk of harm to participants, with the risk determined by the type of activity and participant's profile.

An acknowledgement of risk involves the provider communicating the potential risks for the activities in their programme and an acknowledgement of these risks from the participant or their parent/guardian.

The types of risks for each activity should be communicated to participants and their legal guardians as part of the pre-programme communication (see 4.5.3).

A record of the acknowledgment from participants or their legal guardians shall be obtained by the provider and recorded before the commencement of the programme.

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An acknowledgement of risk shall include at least the following information:

- a) Nature of the risks; or
- b) Information on the possible consequences of those risks.

4.5.6 Participant restrictions

The activity leadership shall prevent a participant from taking part or continuing in an activity if it is determined that the ability of a participant to safely participate is compromised. Conditions under which a participant shall not be permitted to participate in an activity for the safety of themselves or others include, but are not limited to:

- a) displaying symptoms which can be reasonably lead the activity leadership to suspect they can be under the influence of alcohol or drugs that can affect their ability;
- b) taking medication that can affect their ability;
- c) being unable or unwilling to follow instructions;
- d) having inadequate physical ability for the activity;
- e) being ill or injured;
- f) behaving in a manner that is reckless and endangers themselves and/or others;
- g) having a medical condition that can impact on safety that cannot be appropriately managed;
- h) being fatigued to a level that endangers themselves or others; or
- i) not having the necessary equipment such as safety or rescue equipment available.

The activity leadership shall complete an appropriate pre-activity check to ensure participants meet the necessary requirements to undertake the activity.

For conditions other than those listed above, the risk management plan shall be used by the activity leadership to determine what, how and when any restrictions to participate may apply.

Modifications to the activity should be made to address the safety concerns.

4.5.7 Vulnerable participants

When working with vulnerable participants, appropriate policies and procedures to address their needs shall be included as part of the SOPs. This shall include clear directions for reporting suspected abuse or neglect of a vulnerable participant.

Supervision of groups containing vulnerable participants shall be tailored by the provider to meet the needs of both the vulnerable participant(s) and the other participants. This shall include a minimum of one responsible person providing direct supervision of the vulnerable participants.

For vulnerable participants, a confirmation of consent to participate shall be obtained from their parent or legal guardian. Pre-activity information (see 4.5.3) shall be supplied by the provider to a parent or legal guardian before asking for, or gaining, their consent.

4.5.8 Medical conditions

The provider shall have operational procedures in place to identify medical conditions which have the potential to prevent or impair safe participation in OAE activities. This can include, but is not limited to conditions which:

- a) directly affect a participant's ability to participate in the activity;

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- b) affect the participant's health in ways not directly related to participating in the activity;
- c) potentially risk the health and safety of the participant; and
- d) potentially risk the health and safety of others.

Medical conditions can be pre-existing or developed during an activity. Therefore the procedures for identifying medical conditions shall include:

- a) identifying relevant pre-existing medical conditions as part of the pre-programme communications (see 4.5.3);
- b) checking with participants as part of the pre-activity briefing (see 4.5.12);
- c) encouraging participants to be pro-active in reporting possible medical issues;
- d) identifying potential medical conditions during an activity; and
- e) implementing a clear decision-making process for determining if a person is able to safely participate in the activity.

Where a relevant medical condition has been identified in the pre-activity communications, the provider should request the participant to get an appropriately qualified medical practitioner's opinion to determine if the condition prevents safe participation in the activity. The provider should furnish information for the medical practitioner on the context, physical nature and risks involved in the proposed activity when requested.

4.5.9 Other health-related issues

The provider shall have procedures in place to identify any person directly or indirectly involved in an activity, whose physical or mental state can pose a risk to their own or other's safety. This includes, but is not limited to:

- a) apparent intoxication (see 4.5.6);
- b) impaired cognition; and
- c) emotional distress.

Smoking should not be permitted in the activity area.

4.5.10 Privacy of information

Sensitive participant information such as personal medical records shall be stored securely by the provider. Access to sensitive participant information shall be restricted to those who need to know to properly carry out their responsibilities.

4.5.11 Inclusiveness and preventing discrimination and sexual misconduct

The provider shall implement policies to ensure participants are treated fairly and without favouritism, discrimination or sexual misconduct. All employees of the provider, including sub-contractors, who have contact with the participants shall be instructed on appropriate behaviour and the actions to take if inappropriate behaviour is observed in others.

In addition to preventing discrimination, procedures should also be put in place to encourage inclusiveness.

Available access to same-gender responsible persons should be provided to groups with vulnerable participants.

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Areas addressed in discrimination policies and procedures can include but are not limited to:

- a) gender including:
 - pregnancy; and
 - breastfeeding.
- b) gender identity;
- c) sexual orientation;
- d) relationship status;
- e) physical appearance;
- f) race;
- g) cultural sensitivity; and
- h) disability awareness.

In delivering the activities, the provider should have policies and procedures in place to address cultural, religious and gender considerations which can include, but are not limited to:

- a) the impact of dress requirements on activity practicalities and safety;
- b) the impact of attire on participants' modesty;
- c) respect for others' beliefs and practices;
- d) requirements for appropriate social interactions;
- e) requirements for appropriate interpersonal physical contact;
- f) requirements for specific food exclusions; and
- g) requirements for specific cooking arrangements.

4.5.12 Pre-activity briefing

Immediately prior to the commencement of an activity, the activity leadership shall provide a thorough briefing to the participants. This briefing shall include:

- a) the activity tasks;
- b) information about the activity area;
- c) instructions on carrying out the activity;
- d) an introduction of the activity leadership and their roles;
- e) an introduction of supervising responsible persons and their roles;
- f) clear safety instructions;
- g) clear instructions on emergency procedures;
- h) expected participant behaviour;
- i) confirming that participants are agreeable to carrying out the activity;
- j) the process for a participant to cease participation at any appropriate time; and
- k) checking the participants are dressed in appropriate attire.

The briefing should also include information on:

- a) participants' access to food and water;

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- b) safe food handling;
- c) personal hygiene;
- d) how to avoid becoming lost or separated from the group;
- e) what to do if lost or separated;
- f) conservation strategies, including:
 - protecting flora and fauna;
 - removing rubbish; and
 - sanitation; and
- g) appropriate respect for the cultural heritage of the area.

The briefing shall be presented in a way that is appropriate, including accounting for any impairments that participants may have.

4.6 Environment

4.6.1 General

This subclause covers environmental conditions that are generally relevant across many activities. Some activities require more specific measures and/or considerations. These are set out in the respective clauses on specific activities.

When planning for OAE programmes and the individual activities contained within a programme, the provider shall include environmental factors such as the:

- a) operating environment;
- b) immediate surrounding environment; and
- c) wider environment.

The operating environment consists of the artificial and natural spaces in which the activities are located and conducted.

The immediate surrounding environment includes, but is not limited to:

- a) the natural environment;
- b) other people in the immediate area; and
- c) cultural heritage sites in the immediate area.

The wider environment consists of the factors which can affect or be affected by the activity, but are not located or confined to the immediate vicinity of the activity. This includes, but is not limited to:

- a) the weather;
- b) pollution caused by the activity, including:
 - noise pollution;
 - littering; and
 - accidental spills;
- c) pollution which can impact the activity (e.g. haze).

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The provider shall include SOPs in the programme and activity planning which address:

- a) environmental risks (see 4.6.2); and
- b) environmental sustainability.

It is also important that everyone involved in an activity takes responsibility for protecting the natural and cultural heritage of the areas they visit. Annex B provides more information on environmental sustainability procedures.

4.6.2 Managing environmental risks

Environmental factors which shall be included by the provider in the risk management plan (see 4.3) include:

- a) weather and emerging weather variations (check weather forecasts, warnings and associated conditions e.g. heat, wind, rain, lightning);
- b) climatic conditions (e.g. heat, rain saturation, humidity);
- c) pollution (e.g. haze, discarded items, contaminated water); and
- d) activity area's geography – knowledge of the landscape of the activity area and how it relates to environmental conditions including:
 - any available shelter;
 - tree density;
 - ground stability;
 - danger of falling objects (rocks from higher ground, trees, branches, etc.);
 - any exposure to possible lightning;
 - any steep slopes;
 - any health issues specific to that locality (e.g. dengue hot spots); and
 - presence of other people (e.g. park users, villagers, residents) and/or wildlife.

Procedures shall be put in place by the provider to prevent, reduce or address identified environmental risks.

4.6.3 Trigger points

Trigger points are circumstances which indicate a need for an action to be initiated. They can be based on forecast or observed environmental conditions. Identifying trigger points and the actions to be taken in response shall be included in the provider's activity plan. A trigger point can be based on, but is not limited to:

- a) a measured environmental condition including:
 - wind speed;
 - temperature;
 - water level;
 - pollution standards index (PSI) level; and
 - wet-bulb globe temperature level; or
- b) an observed or forecast weather event including:
 - lightning;

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- strong winds;
- flash flooding; and
- fire risk (dry weather).

The provider shall include both forecast and observed conditions as indications of a trigger point being reached in the activity plan.

Actions which can be initiated by a trigger point include, but are not limited to:

- a) changing the frequency of monitoring changing weather conditions;
- b) modifying the activity;
- c) suspending the activity;
- d) terminating the activity; and
- e) evacuating the activity area.

Trigger points in the activity plan shall include all relevant factors and a safety margin.

4.6.4 Weather

4.6.4.1 Weather conditions and associated risks

The impact of weather on an activity depends on:

- a) type of activity being undertaken;
- b) location of the activity; and
- c) available amenities.

Weather conditions which can affect an outdoor activity include:

- a) strong winds;
- b) extreme temperatures;
- c) lightning;
- d) rain; and
- e) drought.

Risks associated with weather conditions include:

- a) unstable or slippery surfaces;
- b) an increased risk of falling branch and trees;
- c) winds inhibiting safe movement;
- d) heat-related illness;
- e) medical conditions related to increased airborne pollution; and
- f) fire.

4.6.4.2 Preparing for weather events

Official weather forecasts and advisories shall be reviewed by the provider and activity leadership:

- a) during the planning of a programme and the activities contained within it;

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- b) immediately prior to commencing an activity; and
- c) regularly, throughout the duration of an activity.

Where on-site observations by the activity leadership indicate an actual or possible risk which is not reflected in a weather forecast, the on-site observations shall take precedence.

Where the weather conditions negatively impact the safe commencement or continuation of an activity, action shall be taken by the activity leadership. See 4.6.3 for actions which can be initiated when a weather trigger point is reached.

Sites which have a heightened level of risk during severe weather events should be predetermined during the activity planning and risk management process (see 4.2 and 4.4). In determining such sites, consideration should be given to the time taken, distance and route to safety when trigger point conditions are met.

4.6.4.3 Hot weather

Where activities are held during hot weather which has not reached the trigger point for cancellation or postponement, the activity leadership shall:

- a) implement regular, appropriate fluid intake to help avoid dehydration;
- b) conduct physically demanding activities under shelter, where possible;
- c) conduct physically demanding activities in a cooler part of the day, where possible;
- d) monitor everyone involved in the activity for heat-related:
 - illness (e.g. heatstroke), and
 - conditions (e.g. heat rash);
- e) institute rest breaks where necessary; and
- f) if required for safety, suspend the activity.

Reasonable steps shall be taken to protect everyone involved in the activity from the adverse effects of sun exposure. The measures can include, but are not limited to:

- a) wearing sun protective clothing which is suitable for the activity;
- b) wearing sun protective headgear which is suitable for the activity;
- c) wearing eyewear suitable for the activity, which protects from UV exposure;
- d) having sunscreen of at least SPF 30+ available;
- e) actively encouraging the use of sunscreen of at least SPF 30+; and
- f) using shaded areas whenever appropriate.

4.6.4.4 Cold weather

Where activities are held during cold weather which has not reached the trigger point for cancellation or postponement, the activity leadership shall:

- a) encourage appropriate choice of clothes to maintain optimal bodily temperature; and
- b) monitor for cold-related illness (e.g. hypothermia).

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4.6.5 Fire danger

Uncontrolled fires are those which are not contained by a fire pit, cleared area or other deliberate fire containment. Such fires can be caused by a natural event (e.g. lightning strike), accident or malicious act.

Factors which increase the risk of an uncontrolled fire include:

- a) type of environment including:
 - type of fuel in the area;
 - amount of fuel in the area; and
 - topography of the area;
- b) climate and seasonal conditions;
- c) current weather conditions;
- d) presence of controlled fires within the area; and
- e) participants' ability to handle the fuel.

The provider shall include in the activity plan the following assessments related to fire danger:

- a) Risk(s);
- b) Likelihood(s); and
- c) Types of impact.

The provider shall establish procedures to minimise the potential for starting accidental fires during activities. These procedures shall include at least the following:

- a) Checking for restrictions on lighting of fires;
- b) Checking before lighting a fire that it is safe to do so in the current weather conditions;
- c) Ensuring a fuel-free perimeter around the fire site;
- d) Extinguishing fires completely when finished; and
- e) Safely disposing any debris.

4.6.6 Tree safety

The provider shall have procedures for managing safety in the vicinity of trees for their planned activities. Such procedures shall include:

- a) checking with the site owner on their tree assessment process;
- b) checking with the site owner for any known tree-based hazards;
- c) carrying out a visual assessment of the trees in the activity area by the activity leadership immediately prior to participants being allowed free access. The visual assessment includes looking for:
 - falling hazards including:
 - unattached branches,
 - dead branches, or
 - heavy fruits;
 - trip hazards (e.g. protruding roots).

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- d) instructing participants to:
- be aware of the potential risks from trees,
 - make basic visual assessment of trees in their immediate vicinity.

Where a tree is identified as posing a potential safety hazard, appropriate mitigation action shall be taken by the activity leadership. This can include:

- a) marking a safety perimeter around the hazard;
- b) informing all involved in the activity of the hazard;
- c) moving the activity to a safer location; or
- d) cancelling the activity.

For activities whose areas cannot be practically inspected, an assessment of the general conditions which can affect tree safety shall be made by the activity leadership. Such conditions include:

- a) recent heavy rain;
- b) forecast heavy rain;
- c) recent strong winds;
- d) current strong winds;
- e) forecast strong winds; or
- f) a known and spreading tree disease.

The results of the assessment and precautions to observe throughout the activity shall be conveyed to the participants by the activity leadership prior to the commencement of the activity.

4.6.7 Drinking water availability and safety

The activity leadership shall ensure that all involved in a programme have ready access to sufficient potable water to meet their needs for the duration of the programme. Potable water can be:

- a) obtained from an on-site source; or
- b) brought to an activity site.

For activities which cover large distances, such as some hiking and kayaking activities, the activity leadership shall ensure that all involved in the activity have a personal water supply which is sufficient to last until the next water refill point is reached.

Where an on-site source for water is used, all involved in the activity shall be informed by the activity leadership of:

- a) the location of the water source; and
- b) any safety considerations regarding its consumption.

Where the available water is not potable or of an unknown quality, a water purification method which makes the water safe for consumption shall be used.

The provider shall ensure that potable water brought to the site is:

- a) of sufficient quantity to serve the needs of all involved in the activity;
- b) be transported in water-tight, food-safe containers; and
- c) be stored in water-tight, food-safe containers.

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4.6.8 Food availability and safety

The provider shall ensure that all involved in their programme have ready access to sufficient food to meet their needs for the duration of the programme. The provider shall establish procedures to ensure food provided is safe and hygienically prepared and served. The food safety measures shall, at a minimum, include:

- a) appropriate storage of food to avoid spoilage;
- b) secure storage to prevent it being accessed by wildlife and/or pests;
- c) appropriate selection of food that can be safely stored for the type, duration and location of the activity;
- d) appropriate selection of food that meets any specific dietary, medical or allergy requirements;
- e) equipment suitable for the hygienic preparation and serving of food;
- f) equipment for cleaning food preparation and serving equipment;
- g) procedures and equipment to ensure the personal hygiene of those preparing and eating food;
- h) procedures and appropriate equipment (including for storage) to prevent cross-contamination of food that can cause an allergic reaction; and
- i) appropriate supervision to ensure safety and hygiene procedures are implemented.

4.6.9 Wildlife

Procedures to minimise the risks associated with wildlife that can be encountered during the activity shall be implemented by the provider, as wildlife encounters can pose a risk to:

- a) participants;
- b) activity leadership; and
- c) wildlife.

To minimise the activity's impact on local wildlife, the provider shall define a geographic area within which the activity is to take place. When defining the boundaries of the activity area, the types of wildlife encounters can include, but are not limited to:

- a) bites and stings from venomous creatures (e.g. bees, hornets, spiders, snakes);
- b) bites and stings from disease-carrying creatures (e.g. Aedes mosquitoes);
- c) confrontations with large animals (e.g. wild boars, crocodiles); and
- d) confrontations with smaller, potentially aggressive animals, especially those travelling in groups (e.g. troop of monkeys, family of otters).

The activity leadership shall inform all involved in the activity of the:

- a) types of wildlife which can be encountered; and
- b) correct behaviour around wildlife, including:
 - observing wildlife from a safe distance; and
 - not feeding wildlife.

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4.7 Equipment and logistics

4.7.1 Equipment requirements and recommendations

The provider shall ensure that all equipment used, whether provided for the activity or brought by a participant or member of the activity leadership, complies with the equipment requirements of this standard.

Equipment for OAE activities can be divided into five general categories. Table 2 provides a list of the categories along with a brief description.

Table 2 – OAE equipment categories

Equipment category	Description
Group	Equipment that is shared among the whole group during the activity.
Leader	Equipment that the activity leadership needs to lead the activity.
Participant	Equipment that participants need to participate in the activity.
Emergency and rescue	Equipment for addressing emergencies or conducting a rescue.
Activity infrastructure	Large scale, built structures e.g. climbing walls and towers, aerial rope courses, zip lines, and artificial caves and tunnels.

Equipment for OAE activities shall be selected by the provider for its suitability to its intended purpose, including being suitable for both the activity and the users. During an activity, the provider shall ensure that sufficient equipment is available for all involved in the activity.

Where possible, spares should be available for the replacement or repair of any equipment damaged during the activity. In general, replacing the equipment should be the first option so as not to impede the activity. Repairs should be carried out after the activity unless they can be done easily and safely during the activity.

The manufacturer's instructions and limitations shall be used by the provider as a minimum standard for the equipment's:

- a) care;
- b) storage; and
- c) use.

Prior to the commencement of an activity, the activity leadership shall check that all equipment for the activity is:

- a) accessible;
- b) suitable for the activity;
- c) in good condition; and
- d) of sufficient quantity.

Where participants are required to provide specific equipment, the provider shall give a checklist of items to them in advance. Prior to the commencement of an activity, the activity leadership shall check that:

- a) each participant has brought all the required items; and

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- b) the equipment brought by participants for the activity is:
 - suitable for the activity, and
 - in good condition.

The provider shall establish a procedure to ensure all activity leaders, assistant activity leaders and participants have access to the equipment required.

4.7.2 Attire and wearables

The activity leadership shall be dressed in appropriate personal attire for the activity and their duties relating to the activity. The provider shall provide the participants with advice on appropriate personal attire for all programme's activities during the pre-programme communication (see 4.5.3). Appropriate personal attire can include but is not limited to:

- a) clothing that:
 - allows sufficient freedom of movement;
 - provides sufficient sun protection; and
 - preserves modesty;
- b) headwear that provides sufficient sun protection;
- c) appropriate footwear; and
- d) methods for safely securing spectacles or sunglasses.

If a participant turns up for an activity without the correct attire, the situation shall be assessed by the activity leadership and corrective action shall be taken if required.

Participants should be reminded to follow these guidelines throughout the duration of the activity.

4.7.3 Communication equipment

The provider shall ensure that the types of communications equipment/system used are appropriate for the context of the activity, the potential hazards and risks involved, and the emergency management plan is in place. This equipment can include, but is not limited to:

- a) whistle;
- b) mobile phone;
- c) satellite phone;
- d) two-way satellite texting device;
- e) two-way radio (e.g. ultra-high frequency radio transceiver);
- f) trunked radio;
- g) EPIRB;
- h) PLB;
- i) tracking device; and
- j) emergency flares.

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4.7.4 First aid equipment

The activity leadership shall ensure that first aid equipment is readily available at all times throughout the course of programme. First aid kits and other first aid equipment shall be:

- a) easily accessible;
- b) appropriate for the activity (see Annex C);
- c) well maintained; and
- d) ready for immediate use.

Any first aid consumables shall be:

- a) within their expiry date (if applicable);
- b) maintained to a minimum stock level; and
- c) appropriately stored.

To determine the first aid and medication requirements for the programme, including which items in what quantities, the provider shall, at a minimum, factor in:

- a) the level of first aid or medical training required for its use;
- b) the size of the group;
- c) the age of the participants;
- d) any pre-existing medical conditions within the group;
- e) the remoteness of the activity;
- f) the ease with which the emergency services can be contacted;
- g) the likely time for emergency services to reach an injured or ill person;
- h) the nature and severity of likely injuries or illnesses associated with the activity; and
- i) the ability to transport and/or carry the equipment.

4.7.5 Use of equipment

The provider shall ensure all equipment is used safely and in accordance with the manufacturer's instructions.

Prior to the use of equipment:

- a) All personnel (activity leadership) and participants shall receive appropriate training.
- b) All equipment shall be checked for serviceability.
- c) Safety equipment with expiry dates (e.g. EPIRP, PLB, flares, etc.) shall be checked that they remain serviceable for the duration of the activity.

When the equipment is being used:

- a) There shall be appropriate supervision for the type of equipment and context of its use.
- b) Appropriate procedures shall be implemented to avoid unnecessary damage to the equipment.

If equipment is damaged in the course of the activity, it shall be withdrawn from use:

- a) immediately, if it poses a safety risk or hazard; or
- b) at the earliest opportunity if there is no immediate risk.

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4.7.6 Maintenance of equipment

Where the provider owns or is responsible for a piece of equipment (i.e. leased equipment, equipment that is stored at the provider's premises), the provider shall put in place procedures to:

- a) maintain the equipment in accordance with the manufacturer's instructions;
- b) implement an equipment maintenance schedule which accounts for:
 - effect of the environment in which it is used;
 - amount of use; and
 - type of use;
- c) identify individual pieces of equipment that require tracking (consumables can be excluded);
- d) track the equipment's condition with inspection and maintenance logs; and
- e) check and remove from service any items that are no longer fit for use.

Where equipment use is not owned or controlled by the provider, the provider shall check with the equipment owner that the equipment is looked after in accordance with the maintenance procedures as described in the list above.

4.7.7 Storage of equipment

Equipment should be stored in accordance with the manufacturer's instructions and recommendations.

4.8 Leadership and professional competencies

4.8.1 General

The programme and activity leadership shall:

- a) possess all required skills, qualifications and experience (see 4.8.3 and 4.8.4);
- b) conduct their activity in a safe and responsible manner;
- c) behave in a respectful and professional manner;
- d) lead by example, supporting and following established rules; and
- e) proactively exclude themselves from an activity or role where they have reason to believe they cannot perform their duties adequately.

4.8.2 Leadership roles and responsibilities

The provider shall ensure that OAE programmes are operated within a clear hierarchical leadership structure, with each person within the structure knowing their job scope and areas of responsibility for the programme.

Terms referring to leadership roles in this standard, e.g. "programme supervisor" and "activity leader", are not intended to directly reflect a specific job title. They merely indicate which level of the hierarchy a statement applies to. OAE providers assign roles to their staff, though their job titles can be different. For example, an OAE provider might assign a staff member whose job title is "head instructor" to be an activity leader. That person's job title remains "head instructor", but they have the responsibility to carry out the requirements of an activity leader as defined in this standard.

Table 3 shows the terms for leadership roles used in this standard.

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Table 3 – OAE programme roles and responsibilities

Role	Responsibilities
Programme supervisor	In charge of an OAE programme including: a) programme schedule plan; b) activity leadership; c) duty personnel and activity changes; d) participants' basic needs; and e) incident and emergency management.
Activity leader	In charge of an activity including: a) safety of all the people involved; and b) learning component.
Assistant activity leader	Assists the activity leader in the safe running of an activity including: a) care of all the people involved; b) learning component; and c) leading and supervising portions of the activity.
Responsible person	Duties within their capabilities and as assigned by the programme supervisor or activity leadership.

Additional levels of hierarchy may be added, however the responsibility for meeting the requirements remains with the designated role. For example, the programme supervisor can have assistants to help them carry out their duties, but the responsibility for the duties being carried out remains with the programme supervisor.

The hierarchical structure for the supervision and management terms used in this standard is shown in Figure 5. In the event of an emergency, the activity leader and/or the programme supervisor shall coordinate with the relevant non-participating contacts.

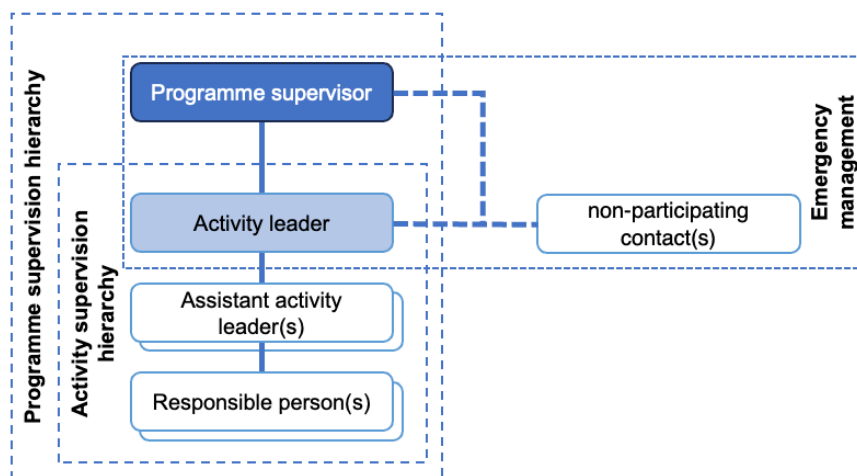


Figure 5 – Supervision hierarchies

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The following scenarios can apply as long as all role requirements are met:

- a) Each programme shall have only one programme supervisor at any point in time.
- b) Each activity shall have only one activity leader at any point in time.
- c) A single person can be both programme supervisor and activity leader.
- d) The activity leadership may consist of one or multiple people.
- e) An entire programme and included activities may be run by a single small team.

Where a person currently in a role (e.g. activity leader) is replaced by another person, a systematic handover of all pertinent information should be performed.

Each person in the leadership team shall be given the authority to make decisions necessary to fulfil their role.

A responsible person's tasks are assigned by the activity leadership. Some examples of the sort of tasks which can be assigned to a responsible person include:

- a) supervising participants not actively participating in an activity;
- b) managing behavioural issues of participants; and
- c) assisting in meeting any gender supervision requirements (see 4.5.11).

Where the responsible person is a representative of the client organisation, they can have duties determined by their own organisation. Some examples of such duties include:

- a) managing the policies and practices of the participants' organisation (e.g. a teacher overseeing school requirements, such as student behaviour standards); and
- b) holding the participants' emergency contact details.

Using these categories as a framework can help identify the type of roles needed for the activity and the appropriate competencies available to lead the activity.

4.8.3 Competencies

The activity leadership shall have the skills and knowledge to lead and carry out the activity safely. This includes, but is not limited to:

- a) Knowledge of the relevant policies and procedures of their organisation including those concerning:
 - activity operations;
 - workplace safety and health;
 - environmental sustainability;
 - participant well-being;
 - equality and diversity;
 - privacy and data protection;
 - communication with colleagues, participants, client organisations; and
 - customer (participant and client organisation) care and service.
- b) Planning OAE activities including:
 - the ability to evaluate participant's information in regard to how it can impact the activity;

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- knowledge of the skills and specialties of the members of activity leadership;
 - knowledge of the activity outcomes and how best to achieve them;
 - familiarity with the activity area(s); and
 - familiarity with using the activity equipment.
- c) Running adventure education activities including:
- conducting an effective pre-activity briefing;
 - providing activity skills instruction;
 - leading an activity group:
 - leading OAE activity sessions;
 - knowing where best to be physically located for supervision;
 - managing group formation;
 - facilitating learning; and
 - managing conflict.
 - identifying hazards;
 - assessing and mitigating dynamic risk during the activity;
 - applying problem-solving skills;
 - minimising environmental impact,
 - interpreting weather conditions in the field,
 - providing basic first aid,
 - providing emergency response.
- d) Concluding an activity including:
- summarising learning to the participants;
 - providing sources for continued learning;
 - correctly packing, storing and returning activity equipment;
 - ensuring the activity area is cleaned; and
 - completing and submitting reports and other paperwork.

The provider shall employ an activity leadership which includes at least one member who has the first aid certification from an accredited organisation.

NOTE – In Singapore, the Singapore Resuscitation and First Aid Council (SRFAC) is the accrediting organisation for CPR and AED certifications.

Additional competencies can be required of the activity leadership based on the type and context of the activity. The provider's activity planning shall identify any additional competencies required for the planned activities. The provider shall ensure the activity leadership includes sufficient members with the identified additional competencies.

All activity leaders and assistant activity leaders should:

- a) undertake professional development to improve their competencies; and
- b) maintain their competencies through regular participation in activities.

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4.8.4 Recognition of competence

4.8.4.1 Recognition framework

The framework used by the provider for recognising competence shall include, in order of priority, at least one of the following. Only when a higher priority assessment method is unavailable shall the next assessment method be used:

- a) Verification of competence level from relevant local governing bodies.
NOTE – Governing bodies for Singapore include the Outdoor Adventure Education Council, outdoor sector bodies and national sports associations.
- b) Approved qualifications from a non-governing entity.
- c) Peer verification.

The types of competencies shall be relevant to:

- a) role being filled (activity leader or assistant activity leader); and
- b) activity which they may be leading.

The factors to be considered for recognition by a training course and by peer verification should include, but are not limited to the following:

- a) Selecting a benchmark competency level based on:
 - national outdoor sector standards;
 - regional outdoor sector standards;
 - national governing body's standards; and
 - established best practices.
- b) Selecting the methods for recognising competency including:
 - demonstrated skills and knowledge;
 - certification of competencies;
 - documented experience; and
 - documented recent practice.
- c) Defining the methods for assessing competency including:
 - using competent accredited assessors; and
 - ensuring ethical behaviour by the assessors.

Annex D provides more information on the recognition of activity leadership competence in Singapore.

4.8.4.2 Competency levels

The provider shall employ an activity leadership with appropriate competence in a range of skills, knowledge, experience, attitudes and attributes along with sound judgment to be able to competently lead and supervise their activity.

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4.8.4.3 Documentation for review

The provider shall have a documented process for recognising and recording the competence of the activity leadership. Copies of this process including the assessments of each member of the activity leadership shall be included in the activity plan documentation (see 4.2.4).

The provider shall ensure that the recognised competencies of various parties are current and relevant by implementing the following procedures:

- a) A review that certifications with an expiry date are valid for the entirety of the programme.
- b) A requirement for a periodic demonstration of practical skills through:
 - performance reviews, and
 - peer reviews.
- c) A review of recent practice as documented in:
 - logbooks, or
 - employee records.

4.8.5 Verification of information provided

The provider shall verify any information provided relating to a member of the activity leadership's competence. The means of verifying information shall include at least one of the following:

- a) Requesting copies of documentary evidence;
- b) Confirming documents are authentic (e.g. by contacting the document issuer, viewing or obtaining certified copies);
- c) Conducting referee checks;
- d) Assessing logged experience;
- e) Assessing previous employee reviews;
- f) Assessing feedback from previous participant; or
- g) Confirming skills assessment(s) through peer recognition or verification.

4.8.6 Supervision of the activity

4.8.6.1 Types of supervision

Supervision by the activity leadership shall be ongoing throughout an activity. The level of supervision required for participants can vary or depend on the context. See Table 4 for an overview of the different types of supervision.

The required level of supervision for each part of a programme and each part of the activities in the programme shall be determined by the provider as part of the programme and activity planning processes (see 4.1 and 4.2). Risk assessments (see 4.3) shall form part of the determination of supervision type.

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Table 4 – Supervision types

Supervision types	Examples of determining factors
Direct supervision	– Activity involves a high level of risk; – Participants require close supervision; or – Activity requires close coordination to run smoothly.
Indirect supervision	– No known high level of risk; – Participants can carry out the activity safely without immediate intervention; or – Activity can run smoothly without immediate intervention.
Remote supervision	– Participants have a sufficient level of competency in the activity; and – Participants are capable of coping with known risks without immediate aid.

4.8.6.2 Direct supervision

Where the plan indicates that direct supervision required, a member of the activity leadership shall remain in close proximity to the participants throughout the duration of the activity.

The level of supervision in an activity may be increased by the activity leadership, based on prevailing conditions including:

- participants' capability;
- increased risk due to environmental conditions; and
- any other factors which the activity leadership determines may require additional supervision.

4.8.6.3 Indirect supervision

Precautions to be taken by the activity leadership when implementing indirect supervision shall include:

- ensuring the participants are sufficiently trained in the activity;
- assessing that the participants are competent in the activity;
- assessing the participants are competent in safe conduct and any emergency procedures;
- ensuring the participants are appropriately equipped;
- establishing clear and understandable boundaries with the participants;
- ensuring there are clear communication lines amongst the participants and activity leadership;
- establishing a recognised point at which the activity is completed; and
- if the activity cannot be safely completed in its current state, having in place clear arrangements to:
 - increase the level of supervision;
 - adjust the activity; or
 - terminate the activity.

4.8.6.4 Remote supervision

Prior to the use of remote supervision, a risk assessment of the activity shall be undertaken by the provider to ensure the activity leadership has the competence and resources to intervene in an appropriate manner if necessary.

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Where remote supervision is used, the provider shall furnish the participant with full details of how the remote supervision is done prior to the start of the activity. Where there is a vulnerable participant in a remotely supervised activity, details of precautionary measures taken shall be provided to the client organisation.

Precautions to take when implementing remote supervision shall include:

- a) ensuring the participants are sufficiently trained in the activity;
- b) assessing that the participants are competent in the activity;
- c) ensuring the participants are competent in safe conduct and any emergency procedures;
- d) ensuring the participants are appropriately equipped;
- e) establishing clear and understandable boundaries for all;
- f) ensuring the participants are given guidance as to the level of difficulty their current pre-activity or existing training is suitable for (e.g. paddling in flat inland water body or in tidal open waters along Singapore's coastline);
- g) ensuring there are clear communication lines between the remote supervisor and those being supervised; and
- h) establishing a recognised point at which the activity is completed.

If the activity cannot be safely completed in its current state, the provider shall ensure that clear arrangements are made to:

- a) increase the level of supervision;
- b) adjust the activity; or
- c) terminate the activity.

Remote supervision of the activity can also include, but is not limited to:

- a) use of communications technology (e.g. CCTV, scheduled radio or phone calls);
- b) use of GPS tracking devices to monitor the participant's progress; and
- c) status updates from the participants at agreed milestones.

4.8.6.5 Additional instructions when implementing indirect or remote supervision

In indirect and remote supervision situations, additional instructions such as, but not limited to, the following should be given:

- a) Start and finish times and locations;
- b) Clear geographical boundaries;
- c) Required or prohibited activities;
- d) Explanation of additional responsibilities placed on those being supervised in this manner;
- e) Where and how the remote supervisor can contact the group;
- f) Where and how the group can contact the remote supervisor;
- g) If the group's progress can be actively monitored;
- h) Any arrangements to pause indirect or remote supervision if this becomes necessary;
- i) Whether details of intentions need to be reported and/or recorded for all or part of the activity; and
- j) Requirements for reporting when starting and finishing.

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4.8.7 Self-care

The activity leadership shall practise self-care and ensure they are not impaired physically or mentally while carrying out their duties.

This shall include:

- a) not being under the influence of alcohol, drugs, medication or any other substance which can impair their physical or mental performance;
- b) managing physical fatigue; and
- c) managing mental fatigue.

Procedures for the management of fatigue shall be implemented by the provider.

These procedures should include:

- a) appropriate allocation of roles and responsibilities;
- b) appropriate scheduling and planning of activity times;
- c) appropriate travel arrangements before and after the adventure activity;
- d) appropriate number and mix of participants to manage;
- e) control measures for specific tasks, jobs and operations;
- f) self-assessment checklists;
- g) reporting fatigue hazards and risks;
- h) managing those who are fatigued;
- i) providing information, instruction, training and supervision; and
- j) monitoring and reviewing procedures and control measures.

4.8.8 Activity group size and supervision ratio

4.8.8.1 Determining group size

The number of participants taking part in an activity at the same time is referred to as the activity group size. The activity group size shall be agreed between the provider and client organisation during the activity planning (see 4.2). The group size can be defined by a maximum and a minimum number of participants. Group size shall be determined to be large enough to perform the activity and small enough that the activity can be supervised effectively.

The provider should use the following factors to determine the group size:

- a) Type of activity undertaken;
- b) Objective of the activity;
- c) Any site-specific restrictions or requirements;
- d) Ability of the whole group to complete the activity in the allotted time;
- e) Participants' level of experience and competence;
- f) Participants' expected dispositions (e.g. behaviours and additional needs);
- g) Potential impact on other users;
- h) Expected and foreseeable environmental conditions;
- i) Quantity and capacity of the activity equipment required (see 4.7);

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- j) Any limits to capacity for emergency response (see 4.4);
- k) Activity leadership's level of competence (see 4.8.3);
- l) Size of (number of people in) the activity leadership;
- m) Acceptable supervision ratio (see 4.8.8.2); and
- n) Activity leadership's level of experience in the environment.

4.8.8.2 Establishing supervision ratio

The supervision ratio for an activity is the ratio of activity supervisors to participants. Activity supervisors can include the activity leadership and responsible persons, where the responsible person has been tasked with supervising participants. Responsible persons accompanying a vulnerable participant shall be included as participants in the activity supervisor-to-participant ratio calculation.

Where applicable, reference should be taken from the respective activity national governing body's recommended supervision ratios and group sizing.

Depending on the type of activity, a single supervision ratio can be applied to the whole activity or individual ratios can be applied to individual portions. For example, checking personal protective equipment (PPE) immediately prior to commencing climbing can require a different level of supervision to monitoring a group waiting their turn. Other factors which can affect the determining of an appropriate supervision ratio include the following:

- a) Group size;
- b) Supervision type(s) (see 4.8.6.1);
- c) Planned duration of the activity;
- d) Activity location;
- e) Availability of access to emergency and logistic support;
- f) Availability of communications; and
- g) Any client organisation's requirements.

Where an activity requires a minimum number of activity leadership members which is more than the agreed supervisor to participant ratio, the minimum number of activity leadership members shall take precedence.

4.8.9 Management of activities

All activities shall be run by the activity leadership according to the corresponding activity plan established by the provider. The activity leadership shall ensure that the activity is well-run according to the pre-determined activity plan. Management of activities by the activity leadership shall include, but is not limited to:

- a) ensuring safety systems are implemented;
- b) correcting known problems or issues as quickly as possible;
- c) ensuring systems and processes are followed;
- d) ensuring that allocated leaders have the required and appropriate competence for their role and responsibilities;
- e) adequately screening participants to ensure suitability for the activity;
- f) ensuring the programme or activity design is suitable for the context in which it is being used; and
- g) planning for contingencies.

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The provider shall establish systems and checks to ensure that any third-party contractor involved in the activity:

- a) adheres to the programme and activity plans;
- b) complies with all client organisations' requirements; and
- c) performs their duties in a professional and safe manner.

4.9 Learning design and facilitation

4.9.1 General

The purpose of learning design and facilitation is for clients and service providers to co-create and monitor learning outcomes, teaching and learning strategies, and the structure of an OAE programme.

OAE activities are conducted with the intent of achieving learning outcomes from any of these three following domains:

- a) Outdoor adventure competencies that support participants in living and interacting in an outdoor environment. The development of these skills can be obtained through indoor and outdoor settings;
- b) Intrapersonal and interpersonal competencies; or
- c) Environmental competencies that enable sensitive interactions with green spaces, urban spaces and local communities and promote conservation and sustainability.

4.9.2 Learning design

Learning design for OAE involves creating, in consultation with the client organisation, a programme to deliver the agreed learning outcomes by:

- a) selecting appropriate activities;
- b) designing the selected activities;
- c) sequencing the activities;
- d) applying teaching and learning strategies; and
- e) assessing the programme upon completion.

Learning design follows a systematic process which provides a logical approach from planning to implementation and evaluation. The evaluation can lead to findings that inform future design decisions.

The learning design process shall be documented by the provider, and the documentation included as part of the programme planning documents (see 4.1).

Where possible, the learning design documentation, or relevant parts of the learning design documentation should be shared with the client organisation to help align the plans with the client organisation's expectations.

Figure 6 illustrates the learning design process.

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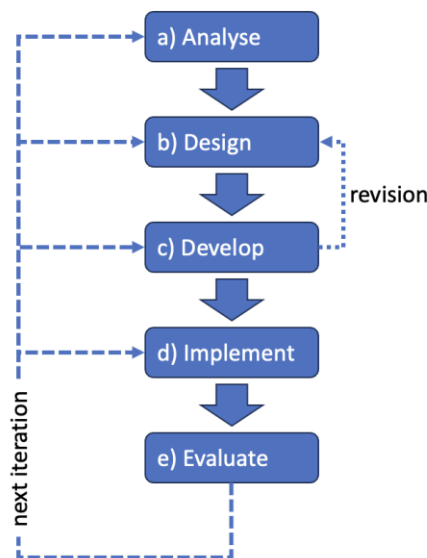


Figure 6 – Learning design process

The following describes each step in more detail:

a) Analyse

An analysis of the OAE programme, carried out by the provider and in consultation with the client organisation, shall include:

- performing a learning needs analysis;
- determining the programme scope; and
- determining the operational scope.

The analysis of learning needs should include but not be limited to:

- the intended learning outcomes;
- general learning outcomes;
- the needs and limitations of the intended participants (see 4.5);
- other stakeholders involved in the programming;
- affordance (i.e. suitability) of the venue; and
- specific manpower requirements.

b) Design

An instructional plan to fulfil the learning outcomes shall be developed by the provider and shall include:

- selecting and sequencing OAE activities based on the learning needs analysis;
- learning objectives;
- instructional strategies;
- instructional methods;
- learning activities;
- instructional material; and
- a contingency plan.

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The plan should take into consideration, but is not limited to, the following:

- A risk management plan (see 4.3);
- An emergency management plan (see 4.4);
- The characteristics of activities that adequately fulfil the learning outcomes;
- The characteristics of the place that informs the possible activities and learning opportunities;
- Facilitation including:
 - matching the abilities of the participants;
 - level of challenge and participation;
 - scaffolding of skills;
 - progression of experience; and
 - sequencing of activities;
- The budget;
- The participants (see 4.5):
 - stages of psychomotor and psychosocial development;
 - stages of group development; and
 - accommodation strategies for participants with special needs and disability;
- The staff:
 - level of competency;
 - agreed ratio;
 - physical ability; and
 - level of fatigue before and during programme;
- The impact of the activity on people and community (see 4.6).

c) Develop

Operating parameters shall be developed by the provider based on the programme design. These parameters should include, but are not limited to:

- selection of suitable activity leadership (see 4.8);
- selection of suitable support staff;
- preparation of the activity leadership and support staff which can include:
 - involvement of stakeholders in the programme;
 - goals of the programme;
 - learning outcomes of the activities;
 - programme schedule;
 - risk and emergency management plans; and
 - medical and learning needs of specific participants;
- selection of the venue which can include:
 - venue rules and regulations;
 - availability of facilities and space for activities; and
 - permits for programme needs;
- provision of amenities;
- operationalisation of the activities by:
 - confirming space for different activities and different portions of the activities;

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- including environmental factors such as weather, climate, wildlife, tidal condition; and
 - checking availability of specific activities;
- catering for meal restrictions requirements;
- logistics; and
- documentation such as:
 - permits;
 - insurance; and
 - SOPs.

d) Implement

The programme shall be implemented by the provider based on the design agreed with the client organisation.

e) Evaluate

The provider shall put in place a process to evaluate the programme including:

- formative and summative feedback in the learning design stages; and
- post programme evaluations (feedback) from:
 - participants;
 - client organisations; and
 - staff.

Based on the evaluations, the programme design can be reviewed for future implementations of similar programmes.

4.9.3 Facilitation

Facilitation is an important part of conducting an OAE activity and is achieved through direct and indirect instructions that guide and support participants in their learning process rather than providing the information. Facilitation for OAE programmes and activities is a planned process which can be adjusted during implementation where necessary.

The provider shall develop facilitation plans for the programme and each activity within it. These plans shall include:

- a) a set of ethics to guide the activity leadership's behaviour to positively influence group and individual behaviours and performance;
- b) policies and procedures for promoting a "minimum impact" approach to using the area so that adverse impacts on the environment and other users are mitigated;
- c) policies and procedures for creating a safe environment where participants are comfortable in sharing and expressing their feelings, thoughts and opinions;
- d) policies and procedures for managing behavioural issues and strong emotional reactions of the participants including referring participants for further help when required; and
- e) policies and procedures for managing conflicts.

Facilitation of activities should support the building of relationships between participants, and between participants and the activity leadership.

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Programme and activity information which supports facilitation of activities shall be communicated by the provider to the participants (see 4.5.3 and 4.5.12). Whenever information is communicated to the participants, they shall be given the opportunity to ask questions and seek clarification.

Techniques to build the participants' self-determination and self-efficacy shall be included in the facilitation plan. These techniques can include, but are not limited to:

- a) offering choices;
- b) building competencies;
- c) encouraging collaborative working;
- d) encouraging self-motivation and motivating others; and
- e) encouraging asking for support.

Facilitation should be supported by a comprehensive knowledge of foundational concepts in education including:

- a) an understanding of human and group development processes;
- b) learning theory;
- c) outdoor education models; and
- d) psychological factors such as:
 - motivation;
 - enjoyment;
 - fear;
 - risk; and
 - stress.

The activity leadership shall continually assess the participants throughout the programme and the individual activities to ascertain their:

- a) ability to meet the activity objectives;
- b) ability to achieve their personal goals;
- c) physical safety;
- d) emotional well-being;
- e) group behaviours; and
- f) physical state.

Based on this assessment, adjustments to an activity shall be made by the activity leadership when necessary to optimally challenge the participants. Such adjustments can include, but are not limited to:

- a) level of participation;
- b) level of challenge;
- c) level of "buy-in" of a participant;
- d) range of choices;
- e) level of support;
- f) sequencing of activity; and
- g) removal of an activity.

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A range of techniques to connect participants' experiences to learning outcomes shall be employed. These techniques can include, but are not limited to:

- a) participants' reflection;
- b) framing of experience;
- c) storytelling;
- d) questioning;
- e) observation;
- f) suggesting experience;
- g) teachable moments; and
- h) transference of learning within and beyond the programme.

Techniques which, along with the general techniques listed above, can help develop a sense of place for participants include:

- a) sensory immersion;
- b) field research;
- c) artifact interviews;
- d) community inquiry;
- e) map-making; and
- f) photo journalism.

The activity leadership shall provide the participants with time and an appropriate environment to reflect on their experiences and articulate their feelings and insights, individually or as a group.

Opportunities shall be provided by the activity leadership for participants to transfer their experiences during the programme to a larger context of school, home, work or other settings.

5 Abseiling and climbing

5.1 General

This clause provides recommendations and requirements specific to abseiling and climbing activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over those in Clause 4.

Abseiling, also known as rappelling, is an activity where the participant descends vertical or near vertical surfaces using ropes and friction devices to manage the descent. Activities which involve abseiling shall employ a belay system.

Climbing activities involve ascending, traversing and then descending vertical or near vertical surfaces. The two types of climbing activities covered in this standard are:

- a) top-rope climbing; and
- b) bouldering.

Activities which are not climbing or abseiling but included as part of the same programme shall be covered by the clause specific to that activity in other clauses of this standard. For example, where abseiling or climbing is included as part of a camping programme, Clause 10 shall cover the camping activity part of the programme.

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5.2 Activity planning

General requirements and recommendations for activity planning can be found in 4.2.

The provider shall include the following as part of the activity planning for an abseiling or climbing activity:

- a) Condition of the course on the activity day:
 - assessing how that can affect the users (e.g. wet course adding difficulty); and
 - having contingency plans in place to adjust to course conditions;
- b) Layout of the course including how it can impact the quality of supervision;
- c) Safe capacity of users for the course; and
- d) Weight limitations of the equipment and course.

The provider shall also include in the activity plan, procedures to:

- a) ensure all personnel wear the correct PPE for their location and current actions, particularly:
 - safety harnesses when there is a risk of falling;
 - helmets when at height; and
 - helmets when at risk from falling objects;
- b) establish a fall zone that is free of obstructions, including debris, people and objects in the fall zone;
- c) establish clearly marked designated waiting areas for participants who are not participating in the activity;
- d) ensure adequate supervision of participants at each stage of the activity (see 5.8.4);
- e) conduct belaying training for participants involved in operating the belay, prior to the commencement of the activity; and
- f) conduct abseil practice for participants prior to the commencement of the activity.

The provider shall have clear safety guidelines and procedures established and incorporated into the activity plan. These guidelines and procedures shall be communicated to both participants and activity leadership by the provider. The safety procedures shall include, but are not limited to having:

- a) at least one member of the activity leadership who is trained and competent in rescue on site for the duration of the activity;
- b) for abseiling activities:
 - a belay system attached to the abseiler (see 5.7.5); and
 - a competent person at the dispatch point who directly supervises the abseil activity and abseilers;
- c) for climbing activities:
 - a belay system attached to the climber (see 5.7.5);
 - a competent person at the dispatch point who directly supervises the climbing activity and climbers; and
 - a spotter present for the first 3 m of a climb;
- d) for bouldering activities:
 - a spotter;

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- an impact absorbing system; or
- a combination of both.

5.3 Risk management

General requirements and recommendations for risk management can be found in 4.3.

In addition to those outlined in 4.3, the potential risks involved in abseiling and climbing activities can include, but are not limited to:

- a) entrapment of:
 - hair;
 - loose clothing; and
 - fingers;
- b) suspension trauma caused by prolonged suspension in safety harness;
- c) collisions;
- d) falling and missing the impact absorption system;
- e) landing causing injury;
- f) distress of participants while at height; and
- g) injury from falling objects.

Procedures established by the provider to minimise the potential harm from falling objects shall include, but are not limited to:

- a) prior to commencing the activity:
 - checking the site for potential falling objects; and
 - where allowed, removing loose objects that are likely to fall;
- b) checking the anchors prior to use;
- c) actively enforcing the use of PPE;
- d) positioning belay areas so that they are not directly below the climber or abseiler, where possible;
- e) designating waiting areas in locations that have reduced exposure to falling objects;
- f) managing groups to minimise time spend within potential fall zones;
- g) minimising movement between areas that are located above other people;
- h) supervising participants while they are located above other people; and
- i) managing spectators and other people moving through the area.

To minimise the possibility of a fall from height, the provider shall implement procedures to:

- a) check the participant's equipment is correctly fitted prior to starting an ascent or descent;
- b) check that participants are correctly attached to the activity ropes and safety systems;
- c) check that ropes used are of a sufficient length for the pitch;
- d) check that all anchor and attachment points within the system are secure and working correctly;
- e) check that unnecessary slack has been removed from the belay system before starting an ascent or descent;

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- f) monitor the continued correct operation of the belay systems;
- g) designate and clearly mark:
 - no-entry areas;
 - waiting areas; and
 - areas that can only be accessed when attached to the belay system; and
- h) anchor the belayer where:
 - the belay system relies substantially on the belayer's weight to arrest a fall; and
 - the weight of the abseiler or climber is greater than that of the belayer.

To reduce the potential for injury to participants or damage to equipment, the provider shall guide participants on how to avoid:

- a) routes where there is potential for the participant to be put in a pendulum swing if control is lost; and
- b) swinging or falling against hard, abrasive and sharp surfaces.

A member of the activity leadership competent in rescue shall be positioned to affect a timely rescue.

5.4 Emergency management

General requirements and recommendations for emergency management can be found in 4.4.

In addition to those defined in 4.4, emergency situations for abseiling and climbing activities can include, but are not limited to:

- a) when a participant hanging in a harness:
 - becomes unconscious;
 - is unresponsive; or
 - is unable to continue to progress;
- b) inclement weather requiring an evacuation of participants; and
- c) equipment failure with an imminent risk of harm.

The provider shall ensure that the use of the rescue systems and procedures in the emergency management plan is practised regularly and at least once annually by the activity leadership members.

The provider shall carry out a review of the rescue procedures and equipment:

- a) at least once every three years;
- b) immediately after the occurrence of an incident;
- c) when there is a change in procedure that requires an update;
- d) when there is a change in the equipment manufacturer's instructions for use;
- e) when there is a change in the rescue system; and
- f) if the equipment shows signs of unexpected damage.

5.5 Participants

General requirements and recommendations for the participants can be found in 4.5.

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In addition to the requirements and recommendations in 4.5.12, the pre-activity briefing to the participants should include but is not limited to:

- a) load limitations for:
 - different sections of the course; and
 - the equipment used;
- b) procedure for terminating and exiting the activity in the event of an emergency;
- c) choices available to the participants regarding the level of challenge;
- d) procedure for the participants' voluntary withdrawal from the activity;
- e) procedure for the PPE check prior to the dispatch or belay;
- f) instructing and/or demonstrating to the participants:
 - self-recovery techniques including resuming the activity;
 - correct climbing and abseiling techniques;
 - fending off the wall when being lowered;
 - actions taken by the belayer to arrest a fall;
 - correct spotting techniques;
 - techniques for descending and falling correctly; and
 - belay/climbing/abseiling calls;
- g) potential hazards and how to avoid dislodging objects;
- h) appropriate action and warnings to give if an object does fall;
- i) method of transferring from the activity belay system to a fixed anchor; and
- j) method of transferring from a fixed anchor to the activity belay system.

5.6 Environment

5.6.1 General

General requirements and recommendations for the environment can be found in 4.6.

5.6.2 Environmental sustainability procedures

Environmental sustainability procedures can include, but are not limited to:

- a) designating the activity and waiting areas in locations that minimise repeated soil and root compaction around trees; and
- b) choosing sites, wherever possible, that have a rocky base and top that can better tolerate foot traffic and the presence of groups.

Further information on environmental sustainability procedures can be found in Annex B.

5.7 Equipment and logistics

5.7.1 General

General requirements and recommendations for equipment and logistics can be found in 4.7.

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5.7.2 Fall protection

The provider shall ensure that fall-protection equipment used for an abseiling or climbing activity is certified as compliant with at least one of the following standards:

- a) UIAA 105;
- b) EN 12277;
- c) ANSI Z359;
- d) ASTM 1772; or
- e) NFPA 1983.

5.7.3 Artificial structure and surfaces

The facility management shall ensure that all artificial climbing, bouldering and abseiling structures:

- a) comply with the requirements described in the EN 12572 series of standards;
- b) have a valid certificate of fitness issued by a professional engineer upon construction or completion of any major modification;
- c) are certified fit for use at least once every 5 years or in accordance with a professional engineer's recommendation if more frequent;
- d) are inspected by a competent person in accordance with the EN 12572 series of standards at least once annually or more regularly in accordance to a qualified person's recommendation;
- e) are periodically maintained in accordance with the recommendations of the facility designer, builder or qualified person; and
- f) have pre-usage inspections in accordance with the instructions of the facility/equipment designers, builders or a qualified person.

The provider shall confirm with the facility management that the above requirements have been met prior to commencing an activity at that facility.

5.7.4 Rescue equipment

Rescue equipment including specialist vertical rescue equipment shall be readily accessible.

Vertical rescue equipment should include but is not limited to:

- a) ascending devices;
- b) belay devices;
- c) connectors;
- d) knives;
- e) pulleys;
- f) prusik loops;
- g) ropes long enough for the longest pitch (applies to abseiling only); and
- h) slings.

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5.7.5 Belay systems

5.7.5.1 Features

Belay systems shall include at least one of the following features:

- a) A top belay system; or
- b) Bottom braking, also referred to as a “fireman’s belay” system.

5.7.5.2 Use

The activity leadership shall ensure that each abseil system comprises at least one belay system to protect the abseiler if they lose control during the descent.

The activity leadership shall ensure that top rope climbing uses a top belay system.

5.7.5.3 Top belay systems

Planning for the use of a top belay system should take into account:

- a) any stretch in the belay system such that it still allows a fall to be effectively arrested; and
- b) likelihood of the abseiler spinning and twisting the belay and abseil ropes together (e.g. free abseiling, abseiling over overhangs).

5.7.5.4 Bottom braking systems

Bottom braking belay systems shall only be used for abseiling activities.

Planning for the use of a bottom braking system should take into account:

- a) any stretch in the belay system such that it still allows a fall to be effectively arrested;
- b) belayer’s competence;
- c) direct supervision of the belayer;
- d) appropriateness of the site including:
 - the ability of the belayer to see the abseiler at all times; and
 - any need to negotiate overhangs;
- e) additional risks when abseilers are free hanging; and
- f) likelihood of objects falling on the belayer.

5.7.6 Safety systems

All persons at height shall be protected from falling with an appropriate safety system.

The activity leadership shall ensure that the safety system for bouldering which is limited to no more than 4.5 m in height includes:

- a) impact absorbing systems;
- b) a spotter providing control of a fall; or
- c) a combination of both.

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5.7.7 Attire and wearables

The provider shall ensure that participants are appropriately attired by checking that:

- a) adjustments are made to the participants' clothing to minimise the chance of entrapment or hindrance of movement during the activity;
- b) participants' hair is appropriately secured to minimise the chance of entanglement or hindrance of movement during the activity;
- c) everyone wears a properly fitted helmet while carrying out the activity or in an area where falling objects are a risk;
- d) participants do not wear hard objects such as hair accessories or jewellery beneath PPE including helmets;
- e) loose jewellery is removed or secured including:
 - bracelets are secured or removed;
 - necklaces are secured or removed;
 - finger rings are removed or taped over; and
 - body piercings are removed or taped over;
- f) participants are wearing appropriate footwear for the activity; and
- g) gloves are worn when abseiling.

5.7.8 Use of equipment

5.7.8.1 Preliminary checks

The provider shall ensure that climbing and abseiling equipment is only used in the manner and for the tasks for which it is designed.

Before use in an activity, the provider shall ensure that all equipment is:

- a) selected to be compatible with other related equipment (e.g. rope diameter is appropriate for the size of fittings used);
- b) suitable for the intended activity;
- c) checked to ensure correct functioning; and
- d) checked for signs of damage or wear.

5.7.8.2 Load-bearing equipment

Load-bearing equipment is designed and expected to support or hold a quantified weight. Such equipment includes, but is not limited to:

- a) ropes;
- b) carabiners; and
- c) climbing harnesses.

The activity leadership shall know the stated strength for any load-bearing equipment, usually provided as the minimum breaking strength (MBS) or safe working load (SWL), prior to use.

The activity leadership shall ensure that the load placed on load-bearing equipment shall not exceed the SWL. Where the SWL needs to be calculated, the manufacturers' instructions shall be followed.

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The provider shall select load-bearing rescue equipment based on the expected loads for a rescue mission with an additional margin of safety.

To ensure that all systems function as intended, the provider shall implement procedures to ensure that, at a minimum:

- a) anchor systems are able to support the intended loads;
- b) all equipment including anchor systems are suitable for the activity;
- c) appropriate knots and connections are used;
- d) checks are conducted to prevent unplanned disconnection of any part of the system; and
- e) attachment to, and disconnection from, the system is carried out correctly.

5.7.8.3 Use of connections

The provider shall periodically review connection methods as well as the equipment and systems used.

When determining connection methods, equipment and systems to use, the following should be factored in:

- a) Type of anchor or anchors being used;
- b) Having sufficient redundancy in case an anchor fails;
- c) What needs to be attached to the harness (e.g. direct connection to rope or a descender device); and
- d) Experience of the person who is connecting the rope or descender device to the harness.

5.7.8.4 Use of harnesses

Connection of harness to rope shall follow the manufacturer's guidelines.

The activity leadership shall ensure that the connection of the belayer's harness to belay devices shall use a locking carabiner.

The activity leadership shall ensure that where participants complete any connection to a harness, it is checked by a competent person.

The activity leadership shall ensure that chest harnesses shall only be used in combination with a sit harness.

Either a combination chest and sit harness or a full body harness should be used:

- a) for very young participants;
- b) when inversion is intended or likely; and
- c) when the security of a sit harness by itself cannot be relied upon due to the participant's
 - level of ability;
 - body shape;
 - body size; or
 - pre-existing condition making a loss of control possible (e.g. epilepsy).

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5.7.8.5 Use of anchors and rigging systems

The activity leadership shall ensure that the activity area includes pre-rigged anchor and belay systems to enable timely and effective rescue responses. Rigging systems for rescue shall be set up to have:

- a) suitable load direction;
- b) suitable load magnitude;
- c) ability to lower the person to safety;
- d) ability to raise the person to safety;
- e) ability to conduct sideways hauls;
- f) ability to perform a contact rescue; and
- g) ability to be deployed and operated swiftly.

The activity leadership shall ensure that dynamic rope shall be used for the belay rope when a fall factor is likely to approach or exceed 0.3.

5.7.9 Maintenance of equipment

5.7.9.1 Equipment records

The provider shall maintain a record of all climbing and abseiling equipment used.

The equipment record should, at a minimum, include:

- a) item individual identifier;
- b) age of the item (e.g. date of manufacture);
- c) date of inspections; and
- d) recommended or maximum lifespan.

5.7.9.2 Inspection regime

Systems subjected to the following conditions shall be inspected regularly:

- a) Repeated, cyclical loads;
- b) Potentially damaging friction; and
- c) Vibration.

The facility owner shall also ensure that all anchors used are periodically inspected by an appropriately qualified person. A record of these inspections shall be maintained. Factors which determine the frequency for the assessment of anchors include:

- a) characteristics of the site;
- b) how many people use the site and how regularly;
- c) applicable manufacturers' recommendations; and
- d) the equipment being used.

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5.7.9.3 Removal from service

In addition to the general requirements described in 4.7.6, factors used to determine the removal of equipment from service shall include, and are not limited to:

- a) type of use;
- b) frequency of use;
- c) prevailing conditions when used;
- d) extreme usage events or patterns (e.g. impacts, 'catches'), if any;
- e) equipment's age; and
- f) length of time the equipment has been in service.

5.8 Leadership and professional competencies

5.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

5.8.2 Competencies

The activity leadership shall have the knowledge and skills to:

- a) set up the activity, including:
 - establishing the belay system; and
 - establishing the rescue system;
- b) instruct participants on the activity, including:
 - wearing of PPE;
 - method for connecting to the activity system;
 - method for climbing or abseiling; and
 - belay techniques;
- c) manage the activity, including:
 - checking participants' connections to the activity system;
 - despatching participants;
 - ensuring that belay ropes are sufficiently taut; and
 - ensuring belayers pay close attention to the climbers and abseilers;
- d) take down the activity after it is completed; and
- e) carry out recovery and rescue operations.

Rescue competencies relating to the activity shall be practised periodically.

5.8.3 Recognition of competence

The activity leadership shall have:

- a) current and appropriate certifications of competence from the relevant local organisation or government body;

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- b) other qualifications which have been approved by the relevant local organisation or government body; or
- c) peer recognised competence as outlined in 4.8.4.

NOTE – In Singapore, the relevant local organisation is the Singapore Sport Climbing and Mountaineering Federation.

See Annex D for more information.

5.8.4 Supervision types

To ensure adequate supervision, the provider shall structure the activity based on the number of competent persons available to take charge of each supervision area. Each qualified person shall be tasked to supervise only one area at a time.

Table 5 indicates the minimum level of supervision which shall be provided for various activity zones.

Table 5 – Minimum supervision types for abseiling and climbing

Zone	Minimum supervision type
Dispatch point	Direct
Landing point	Indirect
Waiting area	Indirect
Belay point	Direct

In addition to the supervision types listed in Table 5, the provider shall ensure that sufficient qualified personnel are available to manage participants in the event of an emergency.

5.8.5 Supervision requirements

5.8.5.1 General

The following factors shall be used by the provider to determine the supervision requirements for climbing and abseiling activities:

- a) Fall protection system used;
- b) Equipment manufacturer's recommendations;
- c) Participants' training progression and competence;
- d) Supervision requirements of participants who are waiting; and
- e) Competence of the person supervising.

5.8.5.2 Supervision for abseiling activities

The activity leadership shall ensure that abseiling activities are supervised as follows:

- a) There is a member of the activity leadership dispatching at the top who also checks the connection of the participant's harness to the abseiling and belay systems.
- b) A member of the activity leadership who is also operating the belay only supervises the lane which they are operating the belay for, unless a top belay is used, in which case, the activity leadership has the option to perform the dispatching role.

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Where the belay is operated by a participant, there shall be:

- a) supervision by a member of the activity leadership;
- b) backup belayers; and
- c) one member of the activity leadership supervising a maximum of 2 such belay systems.

5.8.5.3 Supervision for climbing activities

The activity leadership shall ensure that participants operating a belay in climbing activities are directly supervised.

Additional supervision requirements shall apply for the set-ups described in Table 6:

Table 6 – Supervision requirements for different climbing set-ups

S/N	Set-up	Supervision requirements
1	A member of the activity leadership is operating the belay	A single member of the activity leadership shall supervise 1 participant.
2	Top belay is used with the belayer located at the bottom	– A single member of the activity leadership shall supervise a maximum of 3 belay systems located in close proximity. – A member of the activity leadership with an assistant shall supervise a maximum of 4 belay systems located in close proximity.
3	Top belay is used with the belayer located at the top	– At least 1 member of the activity leadership at the base. – At least 1 member of the activity leadership at the top who shall supervise a maximum of 2 belay systems in close proximity.
4	Activity utilises an auto-belay top-belay system	– 1 member of the activity leadership shall supervise a maximum of 5 belay systems in close proximity. – A member of the activity leadership shall check the participant's connection before allowing the climb.
NOTE – In set-ups 2 and 3, the belayer and backup belayers are participants.		

The activity leadership shall determine the number of climbers that can be supervised while bouldering based on:

- a) competence of the spotters;
- b) type of supervision required based on the ability of the
 - climber(s); and
 - spotter(s);
- c) distance separating climbers; and
- d) site considerations including hazards within the fall zone.

5.8.6 Communication methods

The activity leadership shall employ a system of clear communications—both verbal and non-verbal—to manage the activity.

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Line-of-sight and audible communication should be the preferred means of communicating, where possible.

The activity leadership shall establish an appropriate method of communication between groups at each belay system.

5.8.7 Management of activities

5.8.7.1 Participant management

The activity leadership shall ensure that participants not currently performing the activity are in a waiting area which is located to reduce the likelihood of:

- a) a fall from height;
- b) being struck by a falling object; and
- c) interference with the activity.

Before participants are permitted to operate a belay system or perform spotting duties, the activity leadership shall provide them with appropriate training on their assigned task.

It also should be determined that:

- a) participants are willing to perform the assigned role;
- b) participants are capable of performing their assigned role adequately; and
- c) participants are capable of communicating appropriately with the climber or abseiler.

5.8.7.2 Managing dependencies

When a single member of the activity leadership working independently requires a belayer and relies on participants to belay them, additional procedures that should be considered include:

- a) assessment of the participants' competence to operate the belay system;
- b) use of a backup belayer; and
- c) appropriate selection of the climb route difficulty to minimise the possibility of the activity leader falling while climbing.

5.8.7.3 Management of bouldering activities

Safety considerations specific to bouldering include, but are not limited to:

- a) whether the holds are on:
 - vertical;
 - inclined; or
 - overhanging surfaces;
- b) correct techniques are employed by the participants; and
- c) type of fall-protection used.

5.9 Learning design and facilitation

There are no specific learning design or facilitation considerations for abseiling and climbing activities. General requirements and recommendations for learning design and facilitation can be found in 4.9.

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6 Challenge courses (low and high)

6.1 General

This clause provides recommendations and requirements specific to challenge course activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over the those in Clause 4.

A challenge course is a physical structure comprising either an individual or a series of low and/or high raised elements where a person requires a fall protection system. Zip-lines, whether part of a course or independent, shall be considered challenge courses. Table 7 illustrates the different forms of challenge courses:

Table 7 – Different forms of challenge courses

Physical structure	
Mobile – A portable course that can be set up for use and taken down afterwards.	Permanent – A permanent course is fixed to the ground by any of the various types of foundations, slabs, piers, poles or other means allowed by building codes.
Course elements	
– Low elements, i.e. challenge course elements at a low height, are designed to be operated without a belay system. The appropriate fall safety is achieved through spotting and/or an impact absorbing system. – High elements, i.e challenge course elements at a high height and for which safety cannot be achieved by spotting, require a belay system.	

Activities which are not covered in this clause but are part of the same programme shall be covered by the relevant clause in this standard. For example, where a programme includes both challenge course and camping activities, Clause 10 shall cover the camping activity part of the programme.

Where climbing and abseiling activities are located in the same area as a challenge course activity but are not part of the challenge course, they should be treated as separate activities.

Where challenge courses include integrated climbing and abseiling elements, those elements shall be subject to the requirements and recommendations in Clause 5.

6.2 Activity planning

6.2.1 General

General requirements and recommendations for activity planning can be found in 4.2.

6.2.2 Planning the course

When planning challenge course activities, points the provider shall address include, but are not limited to:

- suitability of the activity site (see 6.6);
- site characteristics such as the:
 - supervision lines of sight; and
 - suitability of waiting areas;

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- c) course element characteristics such as the:
 - difficulty;
 - height; and
 - fall protection system used by the participants;
- d) activity progression incorporating appropriate warm-up activities;
- e) suitable options to allow participants to make choices about their experiences (i.e. “challenge by choice”);
- f) operating procedures and instructions that need to be provided; and
- g) participants’ training required (e.g. spotting, belaying, attaching to safety systems).

6.2.3 Planning the operations

The provider shall include in the activity plans an appropriate pre-activity assessment to determine participants have the necessary pre-requisite skills and knowledge to undertake the activity.

Belay training including climbing calls shall be conducted by the activity leadership prior to the commencement of the activity for participants involved in operating the belay.

At least one member of the activity leadership who is competent in rescue shall be on site at all times during the activity.

The activity leadership shall ensure the fall zone is free of obstructions which have the potential to cause injury to the climber or other participants.

The provider shall identify designated participant waiting area(s) in the activity plans.

6.3 Risk management

6.3.1 General

General requirements and recommendations for risk management can be found in 4.3.

6.3.2 Potential hazards

Potential hazards for challenge courses include, but are not limited to:

- a) operating at height;
- b) improper fitting of PPE;
- c) improper attaching of carabiners and ropes; and
- d) course components and equipment.

6.3.3 Potential risks

Potential risks including those associated with the above identified hazards can include, but are not limited to:

- a) emotional distress of participants while at height;
- b) quality of supervision affected by layout of the course, number of participants and field of vision;
- c) entrapment of hair, loose clothing, fingers in equipment and course components;
- d) prolonged suspension in harness;

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- e) unintended inversion of a participant;
- f) collision between a participant and the course structure;
- g) failure of the impact absorbing system;
- h) failure of the safety equipment, including PPE;
- i) participant falling and injuring other participants;
- j) injury caused by a falling object; and
- k) injury caused by incorrect practices including:
 - improper landing techniques; and
 - improper belaying technique.

6.3.4 Mitigating measures

A helmet shall be worn by all personnel at risk of falling from height or being hurt from any falling objects.

The provider shall provide a fall protection system to be implemented by the activity leadership. A fall protection system consists of safety measures, equipment and procedures that mitigate the risks associated with falls. This shall include at least one of the following:

- a) Spotting;
- b) Impact absorbing system;
- c) Fall restraints; or
- d) Belay system.

An impact absorbing system consists of equipment and associated techniques that cushion the impact of a fall and dissipate the energy, decreasing the risk of injury.

A fall restraint system consists of equipment designed to prevent a person from reaching a fall hazard.

A belay system consists of equipment and associated techniques designed to protect a person carrying out an activity at height by controlling a safety rope attached to the person. The components of a belay system can include:

- a) ropes;
- b) carabiners;
- c) connectors; and
- d) shear reduction devices.

6.4 Emergency management

General requirements and recommendations for emergency management can be found in 4.4.

Emergency events which can occur during a challenge course activity include, but are not limited to:

- a) fall from height which results in serious injury;
- b) participant hanging in a harness for extended period of time;
- c) participant hanging in a harness and is unconscious or unresponsive; and
- d) inclement weather requiring an evacuation of participants; and
- e) equipment failure with an imminent risk of harm.

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The use of rescue systems and procedures in the emergency management plan shall be practised by the activity leadership at least once annually.

The rescue systems and procedures shall be reviewed by the provider:

- a) at least once every three years;
- b) immediately after the occurrence of an incident;
- c) when there is a change in procedure that requires an update;
- d) when there is a change in the equipment manufacturer's instructions for use;
- e) when there is a change in rescue system; or
- f) if the equipment shows signs of unexpected damage.

6.5 Participants

General requirements and recommendations for participants can be found in 4.5.

The provider's pre-activity briefing to participants shall include, in addition to general safety information covered in 4.5.12, information specific to challenge ropes courses as follows:

- a) Any specific instructions regarding course elements;
- b) Maximum number of participants per element;
- c) Working load limits of course elements;
- d) Proper use and handling of equipment;
- e) Termination of activity procedure;
- f) Choice of challenge level (if any);
- g) How to withdraw from the activity;
- h) What PPE to be used;
- i) Safety checks to be completed; and
- j) Actions and behaviours to avoid.

As part of the pre-activity briefing, the activity leadership shall, where applicable, instruct the participants on:

- a) use of PPE including:
 - how to correctly wear the PPE; and
 - how to conduct a PPE check;
- b) operation of the belay system;
- c) all climbing calls used;
- d) spotting a climber;
- e) assisting each other during the activity;
- f) how to fall correctly; and
- g) how to self-recover.

The activity leadership shall supervise the participants practising these skills. The participants shall be assessed as being competent by the activity leadership before being allowed to commence the activity.

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6.6 Environment

There are no specific environmental considerations for challenge course activities. General requirements and recommendations for environment can be found in 4.6.

6.7 Equipment and logistics

General requirements and recommendations for equipment and logistics can be found in 4.7.

The facility management shall ensure that all challenge course structures:

- a) comply with the requirements described in the ANSI/ACCT 03  or EN 15567 series of standards; 
- b) have a valid certificate of fitness issued by a professional engineer upon construction or completion of any major modification;
- c) are certified fit for usage at least once every 5 years or in accordance with a professional engineer's recommendation if more frequent;
- d) are inspected by a competent person in accordance with the ANSI/ACCT 03 or  EN 15567  series of standards at least once annually or more regularly in accordance with qualified person's recommendation;
- e) are periodically maintained in accordance with the recommendations of the facility designer, builder or qualified person; and
- f) have pre-usage inspections in accordance with the instructions of the facility/equipment designers, builders or qualified person.

The provider shall confirm with the facility management that the above requirements have been met prior to commencing an activity at that facility.

The activity leadership shall use a checklist for pre-use inspection that is provided by the designer, manufacturer, installer or qualified person.

6.8 Leadership and professional competencies

6.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

6.8.2 Competencies

The activity leadership shall have the knowledge and skills to:

- a) set up the course for the activity, including:
 - conducting a pre-use check;
 - establishing the belay system; and
 - establishing the rescue system;
- b) instruct participants on the use of the course, including:
 - wearing of the PPE;
 - method for connecting to the course;
 - method for moving across the course elements;
 - method for self-recovery; and

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- use of fall arrest system;
- c) operate the course for the activity, including:
 - checking participants' connections to the activity system;
 - despatching participants;
 - supervising participants' progress through the course; and
 - anticipating hazardous situations for intervention;
- d) take down the course after the activity has completed; and
- e) carry out recovery and rescue operations on the course.

6.8.3 Recognition of competence

The activity leadership shall, at a minimum, be trained and certified:

- a) by accredited bodies for the ANSI/ACCT 03;
- b) in accordance with EN 15567-1 and EN 15567-2; or
- c) in accordance with an equivalent standard endorsed by the local governing authority.

NOTE – In Singapore, the local governing authority is the OAE Council.

See Annex D for more information.

6.8.4 Supervision types

Table 8 outlines the recommended minimum supervision types for challenge courses.

Table 8 – Recommended minimum supervision types for challenge courses

Participant action	Supervision type
Attach and check PPE	Direct
While on the course	Indirect
Waiting participants	Indirect
Participant belayers with top rope belay system (while climber is on the course)	Direct
Continuous belay system (while climber is on the course)	Indirect
NOTE – Some forms of belay systems need direct supervision. Refer to ANSI/ACCT 03 or EN 15567-2 for more information.	

6.8.5 Supervision requirements

The following factors shall be used by the provider to determine the supervision requirements for challenge course activities:

- a) Fall protection system used;
- b) Equipment manufacturer's recommendations;
- c) Participants' training progression and competence;

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- d) Supervision requirements of participants who are waiting; and
- e) Competence of the person supervising.

6.8.6 Supervision ratio

The provider shall comply with the supervision ratio recommended by:

- a) designer of the course;
- b) builder of the course; or
- c) suitably qualified person who has knowledge of the course.

6.8.7 Management of an activity

6.8.7.1 Safety systems

Before permitting climbers to commence an activity, the activity leadership shall conduct a thorough check that:

- a) climbers are wearing their PPE properly; and
- b) if using top rope belay system, the belay teams are ready to belay the climbers.

6.8.7.2 Communication methods

The activity leadership shall employ a system of clear communications—both verbal and non-verbal—to manage the activity.

Line-of-sight and audible communication should be the preferred means of communicating, where possible.

6.8.7.3 Managing dependencies

When a single member of the activity leadership working independently requires a belayer and relies on dependent participants to belay them, additional procedures that should be considered include:

- a) assessment of the participants' competence to operate the belay system;
- b) use of a backup belayer; and
- c) selection of a course section of an appropriate difficulty level to minimise the possibility of the activity leader falling while climbing.

6.9 Learning design and facilitation

There are no specific learning design or facilitation considerations for challenge course activities. General requirements and recommendations for learning design and facilitation can be found in 4.9.

7 Watercraft paddling and rowing

7.1 General

This clause covers the use of watercraft which are controlled and propelled by participants' paddling or rowing on open water and inland water bodies. Watercraft, in this clause, covers both single-occupant and multi-occupant craft including:

- a) canoes;

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- b) kayaks;
- c) stand up paddleboards (SUPs);
- d) rowboats;
- e) dragon boats;
- f) inflatable dinghies;
- g) inflatable rafts; and
- h) other similar watercraft.

This clause provides recommendations and requirements specific to watercraft paddling and rowing activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over those in Clause 4.

For improvised rafting activities, see Clause 8.

Activities which are not covered in this clause but are part of the same programme shall be covered by the relevant clause in this standard. For example, where an activity programme includes both kayaking and hiking then Clause 9 shall cover the hiking activity part of the programme.

7.2 Activity planning

General requirements and recommendations for activity planning can be found in 4.2.

The activity plans for watercraft paddling and rowing activities should include:

- a) a pre-activity assessment of each participant's suitability to take part (e.g. demonstration of water confidence, demonstration of paddling or rowing proficiency);
- b) accounting for all needs in selecting an activity area including:
 - emergency and support craft access (see 7.6);
 - access for people with disabilities; and
 - suitable water conditions for the participants skill level;
- c) selection of appropriate routes (see 7.6) based on:
 - the intended duration and intensity;
 - seasonal factors such as monsoon weather and jellyfish blooms; and
 - the effect of changing water conditions;
- d) identification of alternate activity areas;
- e) identification of contingent routes; and
- f) identification of evacuation and shelter points along the route or within the activity area.

7.3 Risk management

7.3.1 General

General requirements and recommendations for risk management can be found in 4.3.

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7.3.2 Hazards

The risk management plan shall be formulated after factoring in potential hazards such as:

- a) non-seaworthy craft;
- b) unstable craft due to poor loading such as:
 - overloading; and
 - uneven distribution of participants;
- c) uneven terrain while carrying the craft;
- d) paddle board fins;
- e) obstacles in the activity area (e.g. submerged rocks, reefs, etc.);
- f) floating hazards (e.g. fishing nets, shark nets, oil slicks, etc.);
- a) wind strength;
- b) wind direction;
- c) tides;
- d) currents;
- e) waves and swells;
- f) water levels;
- g) other watercraft and users (e.g. ships, motorised and sailing vessels, personal watercraft, kite surfers, divers, snorkellers, etc.);
- h) walking surface for entry, exit or moving about out of the watercraft;
- i) water and environment temperature; and
- j) water contamination.

7.3.3 Risks

The risk management plan shall describe mitigating measures to deal with potential risks, including those associated with the hazards identified above. Risks relating to watercraft paddling and rowing can include, but are not limited to:

- a) craft sinking while in use;
- b) craft capsizing;
- c) drowning;
- d) hypothermia from prolonged immersion in the water;
- e) participants getting trapped in a capsized craft;
- f) injury resulting from an impact between a participant and:
 - craft;
 - paddles; and
 - fins;
- g) tripping while carrying a load;
- h) strain injury from carrying heavy load;
- i) reckless behaviour due to excitement (competition);

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- j) collisions between craft causing damage;
- k) puncture in the membrane of an inflatable craft;
- l) entanglement by leg leash;
- m) loss of paddle; and
- n) participant becoming separated from their craft while in the water.

7.3.4 Mitigating measures

For mitigating measures for the risks identified, refer to the relevant points in Clause 4.

The provider shall identify and document in the risk management plan the threshold for which contingent actions shall be triggered for weather events, such as:

- a) wind strength;
- b) wind direction;
- c) temperature; and
- d) lightning.

Weather-related trigger points shall be based on the national meteorology organisations' weather warnings and information, as well as actual weather conditions.

NOTE – In Singapore, the national meteorology organisation is the Meteorological Service Singapore.

Contingent actions for the trigger points list above can include but are not limited to:

- a) cancelling the activity;
- b) modifying of the activity by:
 - reducing the activity duration; and/or
 - relocating to a more sheltered area; and
- c) evacuating to a safe location.

7.4 Emergency management

General requirements and recommendations for emergency management can be found in 4.4.

In addition to generic emergencies described in 4.4.1, the emergency management plan shall describe procedures to deal with the following emergency situations which can occur during watercraft paddling and rowing activities:

- a) Near drowning incidents;
- b) Participants being out of the activity area and in danger;
- c) Serious injuries (i.e. fractures, head injuries);
- d) Sea creature encounter resulting in envenomation or serious injury;
- e) Missing participant(s);
- f) Instances of heat disorder and injury;
- g) Entrapment and entanglement; and
- h) Mass capsize event.

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7.5 Participants

General requirements and recommendations for participants can be found in 4.5.

Information specific to watercraft paddling and rowing activities in the provider's pre-activity briefing to participants shall include:

- a) description of the activity area and route(s) including:
 - possible aquatic hazards;
 - risks; and
- b) assessment of the water conditions;
- c) correct use of the activity equipment;
- d) a check that participants are wearing appropriate attire (see 7.7.3);
- e) correct fitting of personal floatation devices (PFDs);
- f) when PFDs are to be worn;
- g) correct procedure for a capsize incident;
- h) correct procedure for a "falling out of or off the craft" incident;
- i) hazard recognition;
- j) hazard avoidance techniques;
- k) rescue techniques including:
 - self-rescue; and
 - rescue by activity leaders;
- l) correct handling and navigation of the craft;
- m) navigation requirements when interacting with another watercraft; and
- n) paddler communication within the craft.

Information specific to watercraft paddling and rowing activities provided in the pre-activity briefing to participants should include:

- a) footwear requirements;
- b) storage of personal medication;
- c) securing glasses or sunglasses;
- d) personal thermal protection requirements;
- e) launching and landing procedures;
- f) any designated lead and tail-end craft;
- g) communication between craft;
- h) methods and signals to communicate over longer distance or noisy environments;
- i) appropriate methods for waterproofing equipment;
- j) any requirements relating to associated activities such as:
 - swimming;
 - diving;
 - jumping;

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- swinging or sliding into water; and
- k) safety and communication equipment details.

7.6 Environment

7.6.1 Environmental considerations

General requirements and recommendations for environment can be found in 4.6.

Environmental considerations when selecting locations and times for watercraft paddling and rowing activities should include, but are not limited to:

- a) type and features of the water body used for the activity including:
 - location(s) of a wave or surf zones which can impact the activity;
 - currents (e.g. eddies, overflows and sweep, outlet, channel and structural currents);
 - water depth (e.g. where waves break, reef depth, conditions based on tide, etc.); and
 - water body navigational requirements;
- b) water conditions and/or wave conditions including, but not limited to:
 - tides, tide variances and slack water periods;
 - temperature;
 - quality (e.g. algae bloom);
 - type of wave; and
 - wave height;
- c) type and features of the land areas used for the activity;
- d) presence of other waterborne traffic;
- e) point(s) of entry into the water;
- f) point(s) of exit from the water;
- g) weather conditions including, but not limited to:
 - how they can directly or indirectly affect the activity (e.g. strong crosswinds making keeping to a course more difficult);
 - the potential impact on the health or well-being of the participants (e.g. high heat and humidity, air quality); and
 - the potential for weather events which can place the participants in an immediate risk of harm (e.g. lightning, flash-flooding); and
- h) area closures.

All environmental hazards which have the potential to create a risk to the participants' well-being shall be addressed by the provider in their risk management plan (see 7.3). Such hazards can include, but are not limited to:

- a) fast flowing water;
- b) floating objects;
- c) strainers (i.e. obstructions which participants can be trapped against);
- d) submerged objects; and
- e) water contamination.

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7.6.2 Wildlife

Wildlife encounters can occur both on land and in an aquatic setting for watercraft paddling and rowing activities. The provider's activity plans shall include both types of encounters.

The activity area should be defined to avoid, where possible, all known feeding, breeding and nesting areas for local marine and land-based wildlife.

7.6.3 Environmental protection

When conducting watercraft paddling and rowing activities, procedures that help to minimise damage to the natural environment should be implemented. Procedures specific to watercraft paddling and rowing activities should be practised where possible, include but are not limited to, the following:

- a) A system to secure paddle craft that avoids damage to the bank or vegetation;
- b) Carrying paddle craft and equipment rather than dragging them wherever possible; and
- c) Using established access and egress points.

7.7 Equipment and logistics

7.7.1 General requirements and recommendations

General requirements and recommendations for equipment and logistics can be found in 4.7.

7.7.2 Equipment requirements and recommendations

7.7.2.1 Overview

The provider shall be knowledgeable of the properties and limitations of the activity equipment and ensure it is suitable for the planned activity. Typical equipment for watercraft paddling and rowing activities can include:

- a) watercraft;
- b) paddles or oars to propel the craft;
- c) PFDs;
- d) rescue equipment (see 7.7.2.4);
- e) equipment which aids the supervision of the activity such as:
 - loudhailers;
 - binoculars; and
 - radio communication devices;
- f) equipment which enables the safe conducting of the activity including:
 - appropriate safety craft for the activity; and
 - appropriate rescue equipment for the activity;
- g) equipment to inflate the watercraft (where inflatable watercraft are used); and
- h) signalling devices (e.g. whistles and air-horns).

The provider shall ensure that PFDs, at a minimum, meet the requirements of ISO 12402-5.

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7.7.2.2 Suitability of watercraft

When selecting watercraft for an activity, an assessment of the suitability of its design should be carried out. To be suitable for an activity, the watercraft should:

- a) have the structural strength to withstand a reasonable amount of force within expectations;
- b) have a means of exit without risk of entrapment;
- c) allow exit without risk of entrapment when capsized;
- d) have a stability in use which is suited to the user's capabilities;
- e) be capable of being towed by rope and grasped by hand;
- f) when swamped or capsized:
 - not sink;
 - provide buoyancy to support its crew;
- g) be fitted with appropriate end loops or toggles that do not form entrapment hazards;
- h) the material or finish of the craft does not cause injury;
- i) the colour of the craft is clearly visible to other water users or rescue authorities; and
- j) where integral to the design, the craft has the appropriate fittings (e.g. covers, plugs or hatch covers).

When preparing a watercraft for an activity, it should be checked that:

- a) watercraft is in a good, working condition;
- b) watercraft is equipped with all its required components and accessories, including, where necessary:
 - suitable paddles or oars;
 - appropriate footrests;
 - buoyancy aids;
 - a means for the craft to be secured to prevent floating away;
 - an appropriate means of preventing water entering the cockpit (e.g. spray decks for kayaks);
 - a means of bailing water where appropriate;
 - a means of attaching the user to the craft (e.g. a leg leash for stand-up paddle boards) to prevent separation from the craft; and
 - appropriate navigation light(s).

7.7.2.3 Paddle and oars

The provider shall ensure that the type of paddles and oars used are appropriate for the activity. Parameters which can be used to assess paddles and oars include, but are not limited to the design and dimensions.

7.7.2.4 Rescue equipment

The provider shall ensure that the types of rescue equipment used are appropriate for the activity.

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The rescue equipment should include but is not limited to:

- a) appropriate releasable means of towing a craft (e.g. 15 m of floating rope with a quick release system);
- b) whistle appropriate for water environments;
- c) suitable means of cutting rope;
- d) stirrup to assist users back into their watercraft;
- e) paddle float except where the watercraft are SUPs; and
- f) bailers.

7.7.2.5 Navigational aids and equipment

When used, the provider shall ensure that the type of navigational aids and equipment are appropriate for the activity (e.g. coastal paddling versus island hopping and channel crossing). Typical navigational equipment can include sea charts and a compass.

7.7.3 Attire and wearables

Where a watercraft paddling and rowing activity is being held in open water, the provider shall ensure the participants have a whistle.

7.7.4 Use of equipment

Where possible, paddles should be of a suitable size for the person using them.

The activity leadership shall ensure that the PFD are:

- a) correct size for the person;
- b) correct buoyancy level for:
 - the person; and
 - the environment;
- c) correctly adjusted;
- d) correctly used; and
- e) worn at all appropriate times during the activity.

7.7.5 Maintenance of equipment

Personal equipment should be cleaned after use.

7.8 Leadership and professional competencies

7.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

7.8.2 Competencies

The activity leadership shall be competent in:

- a) watercraft that is used;

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- b) required practices in leading an activity using that type of watercraft;
- c) operating in the environment and water body being used;
- d) appropriate rescue techniques for the watercraft and activity area including lifesaving; and
- e) boat recovery.

Rescue competencies shall be refreshed through periodic practice.

7.8.3 Recognition of competence

The competence of the members of the activity leadership shall be recognised through documented training and certifications as outlined in 4.8.4.

7.8.4 Supervision ratio and group size

The provider shall include the following factors when determining the supervision ratio and group size:

- a) Type of water body being paddled;
- b) Water body conditions;
- c) Access and egress conditions;
- d) Weather conditions;
- e) Visibility;
- f) If access and egress is through surf-like conditions;
- g) Ability of participants' to reliably self-rescue or assisted rescue;
- h) Length and how committing the paddle is to reach a suitable egress location;
- i) Participants' swimming ability;
- j) Size of the watercraft;
- k) Design of the watercraft;
- l) Number of participants in each watercraft;
- m) Suitability of the watercraft for the type of activity and environment;
- n) Suitability of the watercraft for the participants; and
- o) Number of watercraft in use.

When directly supervising a watercraft paddling or rowing activity, the ratio for members of the activity leadership to participants should be no more than 1:20.

For a dragon boating activity, there should be 1 supervising member of the activity leadership in each boat in addition to the steersperson.

7.8.5 Supervision types

7.8.5.1 Factors determining supervision type

The provider shall ensure that the supervision of participants is based on an appropriate risk management assessment (see 7.3).

Members of the activity leadership operating in more remote locations (e.g. further than 1 nautical mile from the coast) should be equipped to manage emergency situations without immediate support from the shore.

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Where an activity requires movement through surf, a minimum of 2 members of the activity leadership shall supervise.

7.8.5.2 Supervision for paddling and rowing activities

Table 9 outlines the recommended supervision types for paddling and rowing activities.

Table 9 – Recommended minimum supervision types for paddling and rowing activities

Open water body/situation	Supervision type
Inland water	Indirect
Inland water “more difficult conditions”	Direct
Coastal waters	Direct
Open waters	Direct

7.8.6 Management of activities

7.8.6.1 Standard procedures

The provider shall ensure that activity management procedures are in place to:

- ensure appropriate communication within the group;
- ensure the ability to communicate to obtain external assistance in the event help is required;
- continually monitor the current weather and water body conditions;
- prevent the group from becoming too spread out;
- ensure correct and safe handling of heavy equipment; and
- ensure appropriate access and egress points are used.

A procedure should be in place to periodically monitor weather forecasts and warnings.

7.8.6.2 Additional procedures

The provider’s activity management procedures should address the possibility of participants undertaking associated activities before, during or after the paddle craft activity. If associated activities (e.g. swimming, diving, swinging/sliding into water, etc.) are allowed or provided by a provider, an appropriate risk management plan shall be implemented.

The provider shall have additional procedures for activities which include transitioning through surf. Such procedures shall include:

- having a path designated for paddling out and paddling in;
- checking the location, terrain, water and wave type is suitable for the activity;
- operating, where possible, away from other water users;
- having participants with the appropriate skills needed including, but not limited to bracing, wet exiting and leaning/edging;
- considering the depth of water where the water breaks;
- establishing appropriate timing for the transition;
- identifying and avoiding obstacles (e.g. rocks, reef, etc.);

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- h) determining activity management system communications; and
- i) securing or stowing all equipment.

7.9 Learning design and facilitation

There are no specific learning design or facilitation considerations for watercraft paddling and rowing activities. General requirements and recommendations for learning design and facilitation can be found in 4.9.

8 Improvised rafting

8.1 General

This clause provides recommendations and requirements specific to improvised rafting activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over those in Clause 4.

Rafts are any buoyant structure which support or enable transportation over water. Improvised rafting is an activity in which a raft is built by participants using available tools and materials, and then used to transport the participants across a body of water. Improvised rafts can be propelled and steered by any of a number of options including:

- a) paddles;
- b) sails; and
- c) motors.

Throughout this clause, the terms “raft” and “rafting” refer to improvised rafts and their use.

Rafts can be constructed by using a combination of materials with a nett buoyancy. In an OAE improvised rafting activity, materials including buoyant items are assembled by the participants, typically by tying or lashing them together, to create the improvised rafts.

Activities which are not covered in this clause but are part of the same programme shall be covered by the relevant clause in this standard. For example, where a programme includes both improvised rafting and canoeing activities then Clause 7 shall cover the canoeing activity part of the programme.

8.2 Activity planning

8.2.1 General planning

General requirements and recommendations for activity planning can be found in 4.2.

The locations for improvised rafting activities shall be selected by the provider to ensure:

- a) construction of rafts is free of hazards; and
- b) appropriate launching and exit points are available (see 8.6.6).

8.2.2 Planning for raft construction

The provider shall include in their activity plan for improvised rafting, provisions to ensure sufficient visibility and lighting for the following:

- a) During the construction of the raft(s);
- b) When launching the raft(s); and

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- c) During on-water use of the raft(s).

The activity shall be scheduled for a time when tidal conditions are expected to be suitable (if applicable).

8.2.3 Safety procedures

The provider shall establish in the activity plans suitable safety procedures according to:

- a) specific attributes of the planned construction site;
- b) attributes of the ingress (launching) and egress (exit) points;
- c) capability of the participants; and
- d) any other factors which can affect the activity from being conducted safely.

Means for all involved in the activity to safely get back to shore in the event of a raft failure shall be detailed in the provider's activity plans.

8.3 Risk management

8.3.1 General

General requirements and recommendations for risk management can be found in 4.3.

8.3.2 Potential hazards

Potential hazards for improvised rafting activities include, but are not limited to:

- a) poorly constructed rafts;
- b) rafts capsizing;
- c) uneven terrain while carrying a raft; and
- d) unseen obstacles in the activity area (e.g. submerged rocks).

8.3.3 Potential risks

8.3.3.1 General risks

Potential risks including those associated with the above identified hazards include, but are not limited to:

- a) raft falling apart while in use;
- b) raft's structural integrity being subjected to stress from propulsion devices such as:
 - engines; or
 - sails;
- c) injuries from:
 - moving raft components;
 - dropped equipment;
 - collision with raft;
 - misuse of paddles; and
 - carrying heavy loads;

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- d) participants becoming entangled in raft components (e.g. ropes);
- e) participants' tripping, particularly while carrying the raft or raft components; and
- f) participants exhibiting risky behaviour due to excitement (e.g. during a competition).

8.3.3.2 Environmental events

The following types of environmental events can pose a risk to the safe running of an improvised rafting activity:

- a) Flash floods;
- b) Lightning; and
- c) Heavy rainfall.

8.3.4 Mitigating measures

Immediately prior to starting the activity, the activity leadership shall review the actual on-site conditions and amend the plans where necessary.

Multiple means for safely getting back to shore can be included in the activity plans. In the event of a raft failure, the activity leadership shall assess the situation and implement the best method(s) for achieving this.

Methods for safely getting back to shore can include:

- a) swimming unassisted;
- b) swimming with assistance;
- c) safety craft to assist; and
- d) safety craft to retrieve and transport.

8.4 Emergency management

General requirements and recommendations for emergency management can be found in 4.4.

Some emergency situations which can occur during an improvised rafting activity include:

- a) near drowning;
- b) out of activity area (in danger);
- c) serious injury (e.g. broken bone, head injury);
- d) envenomation from a sea creature encounter;
- e) lost participant(s);
- f) heat and sun exposure; and
- g) entrapment and entanglement.

8.5 Participants

General requirements and recommendations for participants can be found in 4.5.

The provider shall have procedures in place to check that all participants:

- a) are given clear safety instructions;

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- b) are appropriately attired (see 4.7.2);
- c) are provided with suitable PFDs (see 8.7.2); and
- d) agree to carry out the activity in the given water conditions.

8.6 Environment

8.6.1 General

General requirements and recommendations for environment can be found in 4.6.

8.6.2 Defining an activity area

Improvised rafting for OAE activities shall take place within a defined activity area, as selected by the provider. As long as boundaries are defined and conveyed to participants, the activity area can be in:

- a) open water bodies;
- b) inland water bodies; and
- c) swimming pools.

8.6.3 Location selection

The points listed below shall be factored in when selecting locations and times for improvised rafting:

- a) Type and features of the water body used for the activity;
- b) Type and features of the land areas used for the activity;
- c) Point(s) of entry into the water;
- d) Point(s) of exit from the water;
- e) Weather conditions including, but not limited to:
 - how they can directly or indirectly affect the activity (e.g. strong crosswinds that make keeping to a course more difficult);
 - potential impact to the health or well-being of the participants (e.g. high heat and/or humidity, air quality); and
 - potential for weather events which can place the participants in an immediate risk of harm (e.g. lightning, flash-flooding);
- f) Water conditions including, but not limited to:
 - tides, tide variances and slack water periods;
 - currents (e.g. eddies, overflows and sweep, outlet, channel and structural currents);
 - temperature;
 - depth (e.g. where waves break, conditions based on tide, etc.);
 - quality (e.g. algae bloom and bacteria);
 - type of wave;
 - wave height; and
 - location of a wave zone or surf zone in relation to activity;
- g) water body navigational requirements; and
- h) area closures.

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Points of entry to, and exit from, the water shall be chosen by the provider in the activity plans accounting for factors such as:

- a) water movement such as swash;
- b) surface incline;
- c) presence of abrasive surfaces;
- d) presence of sharp objects including rocks; and
- e) slippery surfaces.

8.6.4 Site assessment

The activity leadership shall conduct an on-site assessment and adjust the points of entry to, and exit from, the water if deemed necessary according to prevailing conditions. The provider shall have a procedure established to guide this on-site assessment.

Based on the prevailing conditions of the day, all environmental hazards which have the potential to create a risk to the participants' well-being shall be addressed by the provider in the risk management plan (see 4.3 and 8.3). Such hazards can include, but are not limited to:

- a) fast flowing water;
- b) floating objects;
- c) strainers (i.e. obstructions which participants can be trapped against);
- d) submerged objects; and
- e) water contamination.

8.6.5 Wildlife

Improvised rafting activities have the potential for wildlife encounters both on land and in an aquatic setting. The provider's activity plans shall include both types of encounters.

The activity area should be defined to avoid, where possible, all known feeding, breeding and nesting areas for local marine and land-based wildlife.

8.6.6 Environmental protection

When conducting improvised rafting activities, procedures that help to minimise damage to the natural environment should be implemented. Procedures specific to improvised rafting include, but are not limited to the following:

- a) Having a system to tie up rafts that avoids damage to the bank or vegetation:
 - carrying rafts and equipment rather than dragging them wherever possible;
 - utilising durable surfaces for:
 - launching the rafts;
 - removing the rafts from the water; and
 - rest locations;
- b) Recovering all raft materials which have become detached in the water;
- c) Retaining and re-using used raft construction materials, where possible; and
- d) Proper disposal of used raft construction materials which cannot be re-used.

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8.7 Equipment and logistics

8.7.1 General

General requirements and recommendations for equipment and logistics can be found in 4.7.

8.7.2 Equipment requirements and recommendations

8.7.2.1 Materials for construction

The provider shall ensure that materials provided for constructing the rafts have properties suitable for the intended activity. Any limitations of these materials shall be accounted for in the provider's activity plan.

Based on the activity goals and participants' capability, the provider should consider the following in the material selection process:

- a) Ease of construction of a suitable raft;
- b) Weight of the constructed raft;
- c) Minimising environmental impact (e.g. non-brittle, re-useable materials);
- d) Risk of injury from the tools;
- e) Risk of injury from the materials; and
- f) Intended degree of difficulty of the entire task.

Typical materials for constructing improvised rafts can include:

- a) for the frame:
 - wooden poles;
 - bamboo poles;
 - PVC pipes; and
 - kayaks;
- b) for buoyancy aid:
 - inner tyre tubes; and
 - plastic barrels;
- c) for lashing the components together:
 - ropes;
 - twine; and
 - straps; and
- d) paddles to propel the raft.

8.7.2.2 Support equipment

The provider shall supply equipment necessary for the safe implementation of the activity. This shall include, but is not limited to:

- a) tools used to construct the rafts;
- b) PFDs that, at a minimum, meet the requirements of ISO 12402-5; and

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- c) equipment which facilitates the safe carrying out of the activity including:
 - appropriate safety craft for the activity;
 - appropriate rescue equipment for the activity;
 - communication devices (e.g. walkie-talkies, mobile phones, and VHF radios); and
 - signalling devices (e.g. whistles and air-horns).

Where the activity is being held in an open water location or large inland water body, the provider shall ensure that each participant has a whistle appropriate for use in aquatic environments.

8.7.2.3 Rescue equipment

Rescue equipment should include but is not limited to:

- a) suitable means of cutting rope;
- b) appropriate means of towing a raft that has a release method (e.g. floating rope with a quick release);
- c) sufficient resources to safely return participants to shore if the raft fails; and
- d) throw-bag or throw-line.

8.7.3 Use of equipment

Assembled rafts shall be checked by a member of the activity leadership before use. This assessment should include checks to ensure that:

- a) lashings, knots and other fixing methods are secure; and
- b) constructed raft appears suitable for the intended task.

The activity leadership shall ensure that PFD used in the activity are:

- a) of the correct size for the user;
- b) correctly adjusted and used; and
- c) worn at appropriate times during the activity.

8.7.4 Maintenance of equipment

Re-usable equipment (including personal items brought by participants) should be cleaned after use.

8.8 Leadership and professional competencies

8.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

8.8.2 Pre-requisites

The activity leadership shall:

- a) wear appropriate personal attire (see 4.7.2);
- b) have access to all necessary equipment (see 8.7.2);
- c) be competent in operating the activity and rescue equipment; and

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- d) have a working knowledge of the operating area for the activity, including:
- hazards in the area;
 - evacuation points; and
 - evacuation routes.

The activity leadership shall be familiar with properties and limitations of the raft construction materials (see 8.7.2.1).

8.8.3 Qualifications and professional competencies

In addition to the competencies listed in 4.8.3, the activity leadership shall, at a minimum, possess skill of water:

- a) lifesaving;
- b) rescue; and
- c) recovery.

Where powered boats are used as rescue craft, the provider shall ensure that the boat driver has a valid boat licence.

The activity leadership shall be competent in the following:

- a) Assessing the water-readiness condition of a raft;
- b) Ropework, knots and lashings relevant to the construction (e.g. clove-hitch, square lashing);
- c) Lifting, flipping technique of a constructed raft; and
- d) Boarding and disembarking from an improvised raft by participants.

Where the activity includes the use of paddles by the participants, the activity leadership shall be competent in the proper use of a paddle including techniques for:

- a) steering;
- b) forward movement;
- c) backward movement;
- d) stopping; and
- e) turning.

The activity leadership shall be competent in safety techniques including:

- a) identifying environmental hazards (e.g. strainers, lightning and tidal conditions); and
- b) managing weight distribution on a raft when:
 - loading;
 - in use; and
 - unloading.

The activity leadership shall be competent in rescue techniques including:

- a) having raft towing techniques;
- b) retrieving participants from a raft;

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- c) managing participants when an improvised raft breaks apart or otherwise fails in the water;
- d) rescuing participants from entrapment (e.g. in raft materials);
- e) using of all rescue equipment;
- f) retrieving participants from the water and onto a powered boat (when a powered boat is used as a rescue craft);
- g) retrieving participants from the water back to shore; and
- h) executing 'TX' and 'kayak to swimmer' rescues (when kayaks are used as safety craft).

8.8.4 Activity supervision

8.8.4.1 Supervision scope

There shall be a sufficient number of activity leaders and assistant activity leaders to safely supervise the activity, taking into account:

- a) number of participants;
- b) number of rafts;
- c) rafts' travel distance over water;
- d) size of the activity area; and
- e) geographic limitations of the activity area.

8.8.4.2 Supervision types and ratio

Supervision shall be based on an appropriate risk management assessment and considerations (see 8.3). Table 10 outlines the recommended supervision types for the activity.

Table 10 – Recommended minimum supervision types for improvised rafting activities

Situation	Supervision type
Raft assembly	Indirect
Launching raft	Direct
Rafting on water	Direct
Beaching raft	Direct
Raft disassembly	Indirect

When supervising directly, the ratio for members of the activity leadership to participants should be no more than 1:20.

8.8.5 Leadership duties

8.8.5.1 General duties

The activity leadership should monitor the construction process of the improvised rafts and guide participants in the design and proper application of knots and lashings, if necessary.

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The activity leadership shall check the rafting areas for any environmental hazards, together with weather forecasts and water conditions. Activities shall not be conducted:

- a) during unsuitable weather including:
 - heavy rain;
 - lightning;
 - strong winds; and
 - reduced visibility; and
- b) where the water conditions are not suitable including:
 - standing waves;
 - boils; and
 - flash floods.

As part of the activity, the activity leadership shall:

- a) conduct water confidence tests with participants, using PFDs in shallow waters prior to first launch;
- b) be aware of participants' swimming abilities and buddy up weak swimmers with stronger ones, where possible;
- c) ensure participants wear their PFDs that fit them. PFDs shall be securely fastened at all times; and
- d) determine appropriate and safe launch sites and landing sites.

8.8.5.2 Pre-activity duties

The activity leadership shall conduct comprehensive pre-activity checks including:

- a) checking the activity area for hazards;
- b) confirming that participants are physically ready to participate;
- c) confirming the activity materials are safe for use, including checks to ensure that:
 - materials used to construct the frame of the improvised raft have no sharp or split edges or surfaces;
 - buoyancy aids to be attached to the improvised raft are not cracked, damaged or leaking; and
 - ropes/straps or material to hold the parts of the improvised raft together are sufficient, not frayed and are adequately strong to hold the raft together;
- d) confirming that the paddle:
 - blades are firmly attached to their shafts; and
 - has no sharp or split surfaces; and
- e) ensuring that the safety equipment is deployed appropriately.

8.8.5.3 Pre-launch duties

The activity leadership shall conduct comprehensive pre-launch checks including:

- a) physical check (e.g. "lift and shake test") to assess the water-readiness of the improvised raft; and

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- b) confirming that loose ends of ropes are tied down and other potential entanglement hazards are removed.

The activity leadership shall brief the participants on:

- a) the risks and the rescue plan including what they are to do in event of:
 - person overboard;
 - drifting;
 - bad weather; and
 - a break-up of the raft;
- b) correct way to mount and dismount from an improvised raft;
- c) warning against dangerous behaviour, such as performing dives or acrobatic acts while they are on the raft;
- d) maintaining directional control of the improvised raft;
- e) safe use of paddles;
- f) safety procedures in the event of a failure of the improvised raft; and
- g) how to safely launch and beach rafts including:
 - not standing within the raft structure; and
 - not being positioned between the raft and beach.

8.8.5.4 Duties during the activity

During the launching and beaching of the rafts, the activity leadership shall:

- a) monitor all launching and beaching of rafts; and
- b) when launching from or landing at pontoons:
 - keep a lookout to ensure participants do not fall into the water and are swept under pontoons; and
 - be aware of the gradient of the beaches/banks and hazards associated with shore-breaks.

Throughout the activity, the activity leadership shall:

- a) keep all participants and their rafts within sight;
- b) maintain quick access to participants and their rafts;
- c) monitor the speed and directions of:
 - tidal currents;
 - winds;
 - floating debris;
 - dangerous marine life; and
 - other watercraft;
- d) direct participants in response to changing situations when necessary; and
- e) instruct participants to immediately make their way towards shore in the event rafts are recalled.

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In the event of a failure (e.g. breaking apart or capsizing) of an improvised raft, the activity leadership shall:

- a) prioritise the safety and well-being of all participants over retrieving equipment;
- b) conduct an immediate headcount;
- c) conduct a preliminary check for injuries;
- d) ensure that no one is entangled by the ropes or otherwise hindered by debris;
- e) ensure all participants are in no immediate risk of harm;
- f) instruct the participants to:
 - swim to shore (if close enough), or
 - await rescue by safety craft;
- g) keep the group members together; and
- h) ensure all participants are removed from the water within a reasonable time.

The safety support requirements detailed in Annex E shall be adhered to throughout the activity.

8.9 Learning design and facilitation

There are no specific learning design or facilitation considerations for improvised rafting activities. General requirements and recommendations for learning design and facilitation can be found in 4.9.

9 Walking, hiking and navigation-based activities

9.1 General

This clause provides recommendations and requirements specific to walking, hiking and navigation-based activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over those in Clause 4.

OAE activities involving participants travelling on foot across natural and built environments are categorised as walking and hiking activities. Trekking, a more strenuous form of hiking, is covered under this clause.

Navigation-based activities involve participants using a variety of tools and techniques to determine their location, then plan and follow a route, either to a set destination or to a series of specific points, usually on foot.

Walking, hiking and navigation-based activities can be single or multi-day activities and can occur across a diverse range of natural and built environments including but not limited to:

- a) rural areas;
- b) wilderness;
- c) coastal areas;
- d) urban and semi-urban areas;
- e) nature reserves and parks; and
- f) park connectors.

An activity can take place in one or across multiple environment types.

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Activities which are part of the same programme but are not walking, hiking or navigation-based activities shall be covered by the relevant clause in this standard. For example, where the activity programme includes camping, Clause 10 shall cover the camping activity part of the programme.

9.2 Activity planning

9.2.1 General

General requirements and recommendations for activity planning can be found in 4.2.

9.2.2 Pre-planning

The following factors should be addressed in the provider's activity plan:

- a) Participants' level of navigational skills;
- b) Availability of appropriate navigational tools;
- c) Intended route;
- d) Availability of suitable sites and means for shelter and/or camping;
- e) Types of terrain which may be encountered during the activity;
- f) Availability of drinking water;
- g) Appropriate access to the start and finish locations;
- h) Suitable emergency access/egress points for the entire activity area;
- i) Hazards and risks associated to the route and entire activity area;
- j) Expected level of difficulty;
- k) Expected weight(s) of equipment that activity leaders and participants need to carry; and
- l) Reasonable access in the initial stages so that participants can withdraw for any reasons.

The provider shall have procedures to reduce the potential of participants becoming separated or lost as part of their activity plan.

9.2.3 Activity and route classification

Classifying activities or specific routes according to their level of difficulty helps to match the participants' capability level with the activity. It also informs the participant of what to expect. The provider shall have a process to assess the level of difficulty of all activities they provide. This process shall include, but is not limited to, the following factors:

- a) Terrain features and characteristics (e.g. undulating terrain, closed primary forest);
- b) Level of physical exertion;
- c) Expected duration of the activity;
- d) Load to be carried; and
- e) Capability of the participants.

9.3 Risk management

There are no specific risk management requirements for walking, hiking and navigation-based activities. General requirements and recommendations for risk management can be found in 4.3.

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9.4 Emergency management

There are no specific emergency management requirements for walking, hiking and navigation-based activities. General requirements and recommendations for emergency management can be found in 4.4.

9.5 Participants

General requirements and recommendations for participants can be found in 4.5.

In addition to the requirements and recommendations in 4.5.12, the pre-activity briefing should include but is not limited to:

- a) expected terrain and environment;
- b) estimated distance, duration or both, of the journey;
- c) estimated weight that a participant is likely to carry;
- d) equipment supplied by the provider; and
- e) equipment, clothing or footwear the participant needs to bring along, including guidance as to weight, size or type of equipment and personal attire.

The activity leadership should also brief the participants on:

- a) what to do if one or more participants are injured, separated or lost;
- b) what to do if one or more of the leadership team is injured or lost;
- c) navigating with the activity navigational equipment (e.g. topographical maps, compasses, electronic devices);
- d) correct use of backpacks (e.g. fitting, adjusting, packing, lifting and carrying);
- e) managing their carried weight;
- f) seeking and building shelters;
- g) outdoor cooking and/or access to food; and
- h) practising field hygiene.

9.6 Environment

9.6.1 General

General requirements and recommendations for environment can be found in 4.6.

9.6.2 General factors

Environmental factors that the participants should be made aware of in walking, hiking and navigation-based activities include, but are not limited to:

- a) terrain (e.g. steep slopes, uneven or unstable footing, cliff edges, etc.);
- b) locality health issues (e.g. dengue hot spots); and
- c) consideration to other people (e.g. park users, villagers, residents).

9.6.3 Water crossings and flooding

The provider shall assess all water crossings along a planned route. A suitable method for safe crossing shall be determined for each crossing. If the crossing cannot be completed with an acceptable degree of safety, it shall not be included in the planned route.

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The assessment of water crossings should include but is not limited to:

- a) risk of being hit and/or swept away by floating or sub-surface debris;
- b) clarity of the water and the visibility of the ground surface of the crossing;
- c) depth of the water at the crossing;
- d) strength of the water flow and the level of effort required to avoid being washed away;
- e) direction of the water flow, particularly the possibility of being swept into dangerous or deadly situations (e.g. into trees in the water, into narrow rock crevices);
- f) makeup of the ground surface of the crossing (e.g. pebbles, sand, small rocks or large rocks) and the risk of foot entrapment and unstable footing during the crossing; and
- g) need to provide additional crossing aids (e.g. walking sticks, guide ropes).

When planning a route for an activity, the provider shall:

- a) avoid the crossing of, or entry into any floodwaters including:
 - active monsoon drains;
 - swollen streams;
 - rivers; and
 - other flooded areas; and
- b) where possible, avoid the crossing of, or entry into areas likely to experience flash flooding.

During an activity, if a crossing or entry into floodwaters becomes necessary (e.g. due to unexpected flooding and with no other paths available), a separate risk assessment shall be carried out by the activity leadership and all risks mitigated to an acceptable level before proceeding.

9.6.4 Wildlife

As walking, hiking and navigation activities can take place over large and varied areas, the provider shall include, as part of their activity plans:

- a) identifying possible wildlife encounters for each different type of area;
- b) establishing procedures for wildlife encounters in less accessible areas; and
- c) establishing a process for noting and reporting possible harm inflicted on wildlife.

9.7 Equipment and logistics

9.7.1 General

General requirements and recommendations for equipment and logistics can be found in 4.7.

9.7.2 Attire and wearables

Personal attire for walking, hiking and navigation activities should, where appropriate, include but is not limited to:

- a) rain gear; and
- b) cold weather wear.

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9.8 Leadership and professional competencies

9.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

9.8.2 Competencies

The activity leadership shall be competent in:

- a) skills which the participants need for the activity (see 9.5); and
- b) demonstrating and teaching those skills to the participants.

The activity leadership shall have the knowledge and skills to:

- a) operate communications systems and equipment;
- b) plan for minimal environmental impact;
- c) plan a travel route in appropriate environments based on the participants' capability;
- d) identify the clothing, footwear and weather gear appropriate to the activity environment and conditions;
- e) pack all needed resources efficiently into a backpack that can be carried ergonomically for the duration of the activity;
- f) set up, maintain and take down a temporary or overnight shelter site;
- g) demonstrate walking skills appropriate to the activity environment;
- h) demonstrate navigational skills appropriate to the activity environment; and
- i) interpret weather information and read local situations.

The activity leadership should have the knowledge and skills to manage the safe crossing of water bodies by all involved in the activity.

9.8.3 Supervision ratio and group size

The provider shall ensure that the following factors are included when determining the supervision ratio and group size:

- a) Field of supervision;
- b) Difficulty of the terrain; and
- c) Type of supervision required (see 9.8.4).

For urban walking and navigation activities in non-remote areas, the provider should control that there is a maximum of 30 participants for every 1 supervising member of the activity leadership where direct supervision is required.

For hiking and navigation activities in remote locations, the provider should control that there is a maximum of 20 participants for every 1 supervising member of the activity leadership where direct supervision is required.

9.8.4 Supervision types

The provider shall establish supervision types based on a risk assessment conducted for the activity (see 9.3). Table 11 outlines the recommended supervision types for the activity.

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Table 11 – Recommended minimum supervision types for walking, hiking and navigation-based activities

Situation	Supervision type
Remote locations	Direct
Non-remote locations	Indirect

9.9 Learning design and facilitation

There are no specific learning design or facilitation considerations for walking hiking and navigation-based activities. General requirements and recommendations for learning design and facilitation can be found in 4.9.

10 Camping

10.1 General

This clause provides recommendations and requirements specific to camping activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over those in Clause 4.

Camping activities for OAE consist of staying overnight in temporary accommodation such as tents, or built accommodation such as chalets or dormitories. Such activities can last for one or several nights. Related activities including cooking and the creation or setting up of shelters. The means by which the participants arrive at the camping site (e.g. hiking or kayaking) are not.

Non-camping activities which are part of the same programme shall be covered by the relevant clause in this standard. For example, where the means used to reach a camping location is by hiking, Clause 9 shall cover the hiking activity part of the programme.

10.2 Activity planning

General requirements and recommendations for activity planning can be found in 4.2.

When planning a camping activity, in addition to the requirements outlined in Clause 4, the provider shall identify all aspects which can affect the welfare or enjoyment of the intended participants and have procedures established to prevent, minimise or otherwise address issues which can arise from them. Such aspects include, but are not limited to:

- a) suitability of the campsite for the participants and type of activity, including:
 - suitability of access to the campsite;
 - suitability of the perimeter security (e.g. fences, defined boundaries, security guards, etc.);
 - availability of medical help including the nearest medical facilities and evacuation route(s);
 - availability of suitable drinking water;
- b) inclement weather and other environmental issues (see 10.6); and
- c) impact of seasonal factors (e.g. monsoon, high temperature periods).

10.3 Risk management

General requirements and recommendations for risk management can be found in 4.3.

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As camping involves overnight stays, often with shared accommodation, there are risks which have a higher potential for occurrence or are unique to camping. The provider shall establish procedures to minimise/mitigate against these risks, which can include, but are not limited to:

- a) sleepwalking;
- b) harassment;
- c) outrage of modesty; and
- d) theft.

10.4 Emergency management

General requirements and recommendations for emergency management can be found in 4.4.

The provider shall establish procedures to manage emergency situations specific to camping activities which include, but are not limited to:

- a) one or multiple missing persons;
- b) outbreak of fire on site; and
- c) mass food poisoning.

10.5 Participants

General requirements and recommendations for participants can be found in 4.5.

In addition to the requirements and recommendations in 4.5.12, the pre-activity briefing should include:

- a) descriptions of the shelter being used;
- b) instructions for correctly setting up the shelter being used, e.g. tents (if applicable);
- c) instructions regarding outdoor cooking;
- d) fire safety;
- e) field hygiene practices;
- f) care and maintenance of equipment; and
- g) adopting environmentally sustainable practices.

10.6 Environment

10.6.1 General

General requirements and recommendations for environment can be found in 4.6.

10.6.2 Fire hazards

The provider shall ensure that there are procedures in place to prevent or mitigate and recover from any incidents of fire. These procedures shall, at a minimum, include:

- a) allocating safe cooking sites away from flammable materials and other potential fire hazards;
- b) preventing heat-related injuries;
- c) means to extinguish or isolate fires;
- d) appropriate storage, handling and transporting of combustible materials; and
- e) evacuation of participants to a safe area.

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10.6.3 Environment-related falling hazards

As part of their activity plan, the provider shall select planned campsites which are safe and suitable for the participants.

When selecting camping and sleeping locations at the planned campsite, the activity leadership shall take steps to minimise the risk of injury from a falling tree, rocks or other hazards. Preventative actions taken when selecting camping locations shall include, but are not limited to:

- a) obeying any official warning signage;
- b) not camping directly under trees that have unattached branches, are heavy fruit bearing trees, or appear to be dead or have dead branches;
- c) not camping near the bottom of cliffs where there is a danger of falling rocks; and
- d) accounting for the potential impact of weather conditions, including severe weather, on the likelihood of objects falling.

10.7 Equipment and logistics

10.7.1 General

General requirements and recommendations for equipment and logistics can be found in 4.7.

10.7.2 Equipment requirements and recommendations

The provider shall have procedures in place to supply suitable camping equipment in adequate quantities for the activity, including:

- a) suitable shelter from the elements;
- b) sufficient sleeping facilities; and
- c) appropriate toilet facilities.

10.7.3 Cooking equipment

Participants shall be trained by the activity leadership on the correct use of the cooking method used, along with any safety procedures associated with the cooking method. Participants who are not directly involved in cooking, but may be present in the immediate vicinity while cooking is done shall be included in this training.

10.8 Leadership and professional competencies

10.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

10.8.2 Competencies

The activity leadership shall have the knowledge and skills to lead a camping activity in the activity environment. This shall include implementing and teaching the following:

- a) Minimal environmental impact camping;
- b) Selection of camp and shelter locations that are appropriate for the weather conditions;
- c) Identification of the clothing, including footwear and weather gear appropriate to the camping environment and conditions;

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- d) Managing the resources (e.g. food, water, shelter) needed to sustain participants overnight for the duration of the activity;
- e) Setting up, maintaining and taking down a temporary shelter site;
- f) Operating and maintaining the cooking equipment used for the activity; and
- g) Reading and understanding weather information.

The activity leadership shall also have the knowledge and skills to demonstrate, teach and implement the following:

- a) Fire handling skills appropriate to the activity environment;
- b) Food preparation and cooking skills appropriate to the activity environment; and
- c) Personal hygiene and hygienic food handling practices in the activity environment.

10.8.3 Supervision ratio and group size

When determining the activity group size, the provider shall take into account:

- a) building capacity (when a building is the campsite);
- b) load capacity of floors (where applicable);
- c) availability of basic amenities (e.g. toilets); and
- d) availability of communal space.

When determining the supervision ratio, the provider shall take into account:

- a) the equipment, method and arrangements factoring in the:
 - number of participants involved in cooking;
 - number participants using or in the vicinity of a naked fire for any purpose (if any); and
 - the group composition and sleeping arrangements.
- b) the competence of the participants including:
 - previously taught skills;
 - assessed skills; and
- c) the ability to provide suitable emergency management.

For general camping activities which only require indirect supervision, there should be a maximum of 30 participants for every 1 supervising member of the activity leadership.

For cooking activities, there should be a maximum of 15 participants for every 1 supervising member of the activity leadership when direct supervision is required and a maximum of 20 participants for every 1 supervising member of the activity leadership for indirect supervision.

10.8.4 Supervision types

Supervision shall be based on an appropriate risk management assessment and considerations (see 10.3). Table 12 outlines the recommended supervision types for the activity.

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Table 12 – Recommended minimum supervision types for camping activities

Situation	Minimum supervision type
Assembling and disassembling tents	Indirect
Use of any fire	Direct
Handling combustible fuels	Direct
Food preparation and cooking	Indirect
Sleeping in shelters	Indirect

10.8.5 Management of activities

A member of the activity leadership shall be contactable throughout the activity, including during the camp silent hours (i.e. late night and early morning hours of the activity when most are asleep). The participants shall be informed how to contact the activity leadership or a responsible person during the camp silent hours.

There shall be sufficient activity leadership and responsible persons to manage large-scale incidents throughout the entire activity, including when indirectly supervising the participants during the night. Example of large-scale incidents include a fire breaking out at the activity location, or an extreme weather event.

The provider shall have a procedure to check that participants are present in their respective assigned shelters during the night.

10.9 Learning design and facilitation

There are no specific learning design or facilitation considerations for camping activities. General requirements and recommendations for learning design and facilitation can be found in 4.9.

11 Experiential learning initiatives and games (ELIG)

11.1 General

This clause provides recommendations and requirements specific to ELIG activities. They are to be used in conjunction with the recommendations and requirements in Clause 4. The provisions detailed in this clause shall take precedence over those in Clause 4.

ELIG activities are individual or group-based activities typically conducted at ground or low level, at indoor or outdoors locations, and do not require special purpose infrastructure. However, some ELIG activities can involve participants handling activity props and equipment.

ELIG activities include activities such as ice-breakers, energisers, de-inhibitors, trust and support activities, problem-solving games. ELIG activities are distinct from low element challenge course activities (Clause 6).

Activities which are not covered in this clause but are part of the same programme shall be covered by the relevant clause in this standard. For example, where a programme includes both ELIG and camping activities then Clause 10 shall cover the camping activity part of the programme.

11.2 Activity planning

General requirements and recommendations for activity planning can be found in 4.2.

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The provider shall sequence the challenges in the activity to facilitate assessment of the participants' ability to undertake challenges, starting with lower risk activities before proceeding to challenges with a higher risk. The participants should be assessed on their:

- a) mental readiness;
- b) fitness level;
- c) mobility; and
- d) behavioural reliability (adherence to expected behaviour).

Factors that can affect activity planning include:

- a) restrictions to physical capability introduced as part of the activity (e.g. being blindfolded, tying of legs together, being muted, carrying of equipment);
- b) environmental factors (e.g. visibility, ground gradient) ;
- c) type and weight of activity props used; and
- d) demographics (e.g. primary school students vs older youth);
- e) participants bearing the weight of another person;
- f) use of confined spaces; and
- g) proximity of hard infrastructure such as fences, walls, gates, etc.

11.3 Risk management

General requirements and recommendations for risk management can be found in 4.3.

ELIG activities which involve potential risks can include, but are not limited to:

- a) walking quickly or running between locations;
- b) lifting or supporting a person;
- c) lifting, carrying, pushing or passing an activity prop;
- d) stepping or jumping on or off an activity prop;
- e) throwing an activity prop;
- f) crawling or sliding between locations;
- g) walking or moving while blindfolded; and
- h) holding one another using hands and other parts of body.

Potential hazards for ELIG activities include, but are not limited to:

- a) debris in a fall zone including;
 - litter;
 - leaves, twigs, etc.; and
 - participants' belongings;
- b) uneven ground;
- c) hard ground;
- d) hard natural elements like boulders, trees;
- e) hard activity structures and props like poles and planks;

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- f) slippery or loose surfaces;
- g) personal belonging like bangles, digital devices, bulky items;
- h) confined spaces;
- i) pre-existing medical conditions; and
- j) extreme weather conditions.

Potential risks including those associated with the above identified hazards can include, but are not limited to:

- a) slip, trip, fall incidents;
- b) collisions between participants;
- c) participants colliding with objects;
- d) pinching injuries;
- e) abrasion injuries;
- f) negative interpersonal interactions (physical, verbal);
- g) triggering a pre-existing medical condition;
- h) worsening a pre-existing medical condition;
- i) heat injuries;
- j) dropped objects; and
- k) projectiles.

11.4 Emergency management

General requirements and recommendations for emergency management can be found in 4.4.

The provider shall establish procedures to manage emergency situations which can occur during an ELIG activity, which include, but are not limited to:

- a) serious injuries; and
- b) emotional trauma.

11.5 Participants

11.5.1 General

General requirements and recommendations for participants can be found in 4.5.

11.5.2 Pre-activity briefing

The provider shall have procedures in place to ensure that all participants:

- a) are given options to determine how they want to participate in the activity;
- b) are able to adjust their participation in the challenge (including opting out) at any time of the activity if they do not feel safe or are not comfortable;
- c) are able to seek help when required; and
- d) are informed of the appropriate places and methods for safe physical contact for ELIGs where there is a need for close physical contact amongst participants.

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In addition to the pre-activity briefing, the provider should give instructions and explanations at appropriate junctures.

11.6 Environment

General requirements and recommendations for environment can be found in 4.6.

Environmental considerations for ELIGs include, but are not limited to:

- a) space appropriate for the conduct of activity;
- b) type of the ground surface;
- c) condition of ground surface;
- d) boundaries and no-go zones;
- e) availability of shelter; and
- f) weather.

Procedures for sustaining the environment, which are specific to ELIG should include but are not limited to:

- a) minimising damage to the environment including:
 - rotating between challenge locations to avoid excessive wear on the vegetation; and
 - utilising durable surfaces, wherever possible, e.g. for rest locations;
- b) recovering all activity materials;
- c) retaining used activity materials which can be re-used; and
- d) proper disposal of activity materials which cannot be re-used.

11.7 Equipment and logistics

General requirements and recommendations for equipment and logistics can be found in 4.7.

The provider shall proceed with the activity only when participants are appropriately attired. The following factors shall be used to determine suitable attire for an activity:

- a) Type of movement (e.g. slow walk vs running);
- b) Environmental considerations (e.g. beachfront vs a multi-purpose hall);
- c) Visibility;
- d) Lighting in dark conditions; and
- e) Likelihood of skin injury.

The provider shall ensure participants wear appropriate clothing and coverings to minimise the risk of injuries from the nature of the activity or the activity location, including:

- a) blisters;
- b) cuts;
- c) abrasions; and
- d) contusions.

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11.8 Leadership and professional competencies

11.8.1 General

General requirements and recommendations for leadership and professional competencies can be found in 4.8.

11.8.2 Competencies overview

The activity leadership shall have a clear understanding of the following aspects of each initiative or game:

- a) Risks;
- b) Difficulty level;
- c) Safety instructions;
- d) Set-up;
- e) Rules;
- f) Variations due to outcome or safety; and
- g) Desired learning outcome(s).

11.8.3 Supervision types

The provider shall base supervision types on an appropriate risk management assessment (see 11.3) Table 13 outlines the recommended supervision types for the activity.

Table 13 – Recommended minimum supervision types for ELIG

Situation where participant(s) are involved in:	Supervision type
Lifting and supporting a person or a heavy prop	Direct
Moving about while blindfolded	Direct
Moving about in conditions involving limited visibility	Indirect
Fast movement (e.g. running)	Indirect
Throwing a hard or sharp object	Direct

11.8.4 Supervision ratio

Where direct supervision is required, the ratio for a member of the activity leadership to participants should be no more than 1:20.

11.8.5 Management of activities

The activity leadership shall be positioned or given the means to intervene directly in an activity and to provide additional support if required.

Participants who are not actively involved in a challenge should be appropriately engaged and supervised so that they do not interfere with the challenge(s).

The provider shall have procedures in place to manage any spectators, incidental passers-by or other people in the vicinity of the activity who are at risk from the activity or present a risk for the participants.

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Special attention should be paid to modesty and cultural sensitivity where the activity is likely to create close physical contact amongst participants. The provider shall instruct participants on the boundaries for physical contact and methods for protecting their modesty during an activity.

The activity leadership shall teach participants techniques for spotting one another where activities involve participants:

- a) being lifted or carried;
- b) moving while blindfolded; and
- c) being otherwise at risk of a fall which can cause injury.

The activity leadership shall teach and demonstrate techniques for correct:

- a) lowering of fellow participants,
- b) being lowered,
- c) falling; and
- d) landing.

The activity leadership should define the limits for each initiative or game area for the participants.

The activity leadership should inform the participants of the appropriate social boundaries for the initiative or game.

11.9 Learning design and facilitation

General requirements and recommendations for learning design and facilitation can be found in 4.9.

Additionally, the activity leadership should prepare variations to a planned learning design and facilitation activity so as to maintain the participants' engagement as the activity progresses.

Annex A (informative)

Medical conditions to factor into risk management

A.1 Medical conditions which can affect participation

Some medical conditions can either prevent or limit a person's involvement in an OAE activity. The degree to which an individual with a pre-existing medical condition can participate depends on the nature and severity of their medical condition.

NOTE – A person in this case refers to anyone involved in an activity including the activity leadership.

Pre-existing medical conditions which can affect participation should be individually assessed as part of the risk management planning. Such conditions include:

- a) allergies, especially those which can result in anaphylaxis (see A.3);
- b) asthma;
- c) epilepsy;
- d) diabetes;
- e) heart conditions;
- f) blood conditions that can affect bleeding or clotting;
- g) mental health issues including those which may result in:
 - impaired memory;
 - altered memory;
 - impaired reaction times; or
 - mood changes;
- h) infectious diseases (e.g. gastroenteritis);
- i) physical disabilities including:
 - impaired sight;
 - impaired hearing;
 - impaired movement; and
 - impaired dexterity;
- j) pregnancy;
- k) physical injuries; and
- l) any other medical conditions which result in:
 - abnormal blood pressure;
 - fainting;
 - dizziness;
 - fatigue;
 - muscular/skeletal complaints; or
 - migraines.

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Medical conditions can also develop or only become apparent during, or just prior to, an activity. The risk management planning should include processes for identifying, evaluating and appropriately managing such situations. Medical conditions which can develop during a programme and can affect participation in an activity include:

- a) infectious diseases (e.g. gastroenteritis);
- b) reactions or side effects to medications; and
- c) physical injuries.

A.2 Managing medical conditions

Risk management planning for medical conditions should include what measures can be implemented to enable a person to participate while maintaining a safe environment for all involved.

Pre-existing medical conditions should be a determining factor when planning:

- a) activity supervision types;
- b) group sizes;
- c) refreshments;
- d) rest time(s);
- e) first aid provisions; and
- f) emergency management procedures.

The planning for the above should also include procedures to follow when a medical condition develops or becomes apparent during or just prior to an activity.

A.3 Allergy and anaphylaxis safety

Anaphylaxis, an acute reaction to an antigen, is a medical incident with potentially serious consequences. As such a risk management plan should include steps to mitigating any occurrence, including the following:

- a) Having allergy management plans in place;
- b) Having an appropriate level of awareness and training for the activity leadership to:
 - manage allergies, and
 - provide first aid treatment; and
- c) Excluding foods that have a high prevalence of allergies in the community (e.g. peanuts).

A.4 Health and well-being

The following questions can help when considering developing risk management plans and procedures to address issues relating to participants' health and well-being:

- a) Have you written the essential eligibility criteria for your activity/programme?

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- b) Do your participants know what is expected of them physically and mentally?
 - Are participants informed of this, is this in your pre-programme information and how is it portrayed in your marketing?
- c) Do you have a medical advisor or access to one?
- d) If a doctor is required to complete a participant health form, do they understand the extent and rigour of the activities in the programme?
 - Do you provide them with information to improve this understanding?
- e) If you receive health information, who reviews the information?
 - How do they decide what to do with the information?
- f) Who makes the ultimate decision about participants – the provider, activity leadership, participant's doctor or your medical advisor?
- g) What tools does the person have, to make these decisions?
- h) Can a decision-making document that indicates the type of considerations and what level of participation is appropriate to be useful? (see Table A.1)
- i) How and when do you share health information with the activity leaders and assistant activity leaders?
- j) Is there a mechanism for participants to update you on new or changed medical conditions?
 - Does this also extend to people who need to know about the change?

Table A.1 – Example of a decision-making document

Assessment for suitability (Participant)	
Medical condition:	
Does this condition prevent participation in the activity? (Yes/No/Conditional)	
If "Conditional", list the conditions for participation: (e.g. limits to the participants' physical capability; the need for a carer; the need for qualified medical support)	
Can all the listed conditions be fully met? (Yes/No)	
If "No", list the exceptions:	

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Annex B (informative)

Environment sustainability procedures

B.1 Minimising impact

Human activities can impact the environment in many ways, not all of which are immediately apparent. Minimising the impact on the environment helps to limit the possibility of any negative effects. Procedures to achieve this can include:

- a) using durable surfaces for travel and camping;
- b) appropriate disposal of waste;
- c) removing only litter from the area;
- d) controlled use of fire;
- e) respecting wildlife; and
- f) showing consideration for others.

B.2 Minimising the impact of OAE activities

Activities held in natural environments typically involve a concentration of people in an area. To minimise the impact this can have on the environment, procedures should be established to:

- a) use durable surfaces, where possible (see B.3);
- b) choose activity locations which can have a minimum effect on the natural vegetation;
- c) properly dispose of waste (see B.4);
- d) minimise human impact (see B.5);
 - remove, where feasible, rubbish found in the area; and
 - avoid disturbing or modifying vegetation and other natural features;
- e) respect wildlife (see B.7);
- f) show consideration to hosts and other users (see B.8);
- g) use only biodegradable hygiene products, and sparingly;
- h) use established facilities rather than temporary ones where possible; and
- i) where possible, leave the activity site in a better condition than it was found.

Additional procedures should be established for camping activities, as they involve the concentration of a group of people in a confined area for an extended time, usually at least overnight. To minimise the impact this can have on the environment, and in addition to that listed above, procedures should be established to:

- a) use no more space than is necessary for the campsite;
- b) disperse campsite activity in natural areas to avoid creating focused areas of impact;
- c) locate campsites so as to avoid negatively impacting any water sources (e.g. creeks, canals, reservoirs);

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- d) minimise the impact of campfires (see B.6);
- e) avoid disturbing or modifying vegetation and natural features around campsites;
- f) use floored tents with poles;
- g) not dig trenches around tents; and
- h) where possible, leave the campsite in a better condition than which it was found.

B.3 Protecting ground surfaces

When planning and carrying out an activity, use durable surfaces wherever possible. Durable surfaces include:

- a) established tracks;
- b) established campsites;
- c) gravel; and
- d) rock.

When travelling as part of an activity:

- a) use existing tracks, pathways and roads, do not establish new tracks;
- b) walk in small parties to minimise damage; and
- c) walk in single file when on narrow paths to avoid widening the track.

B.4 Dispose of waste properly

All waste should be disposed of correctly. Some biodegradable waste may be allowed to be disposed of on-site, but most waste should be either removed or placed in suitable on-site bins. Waste disposal can include (but is not limited to):

- a) carrying out and disposing of all food, organic and general waste appropriately;
- b) separating recyclable waste, where possible, and carrying out and disposing of appropriately;
- c) burying or burning food waste only where permitted by the land owner or land manager;
- d) disposing of waste water by scattering a suitable distance from any water source after straining out any food and solid matter (strained food waste is disposed of as described above); and
- e) checking tracks, activity sites and campsites are free of spilled food waste prior to departure.

The correct disposal of human waste and hygiene includes:

- a) using existing established toilets for human waste, where possible;
- b) where no toilets are available, either carrying human waste out in a suitable container to be disposed of later or, if allowed in the area, disposing of human waste in a temporary 'pit toilet';
- c) constructing a pit toilet (when appropriate) at a suitable distance from watercourses to avoid contaminating the water and campsites (e.g. dig the pit or hole, 8 – 15 cm deep in the biologically active layer in the soil and refill the hole after toileting);
- d) carrying out personal hygiene waste in a suitable container to be disposed of later; and
- e) burial or burning of personal hygiene waste only conducted where permitted by the land owner or land manager.

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B.5 Minimising human impact

With the limited natural resources in Singapore, and the relatively high usage of most activity areas, it is important that every effort be made to minimise the impact activities have on the area. A key part of this is not removing or altering any part of the natural environment. This includes:

- a) ensuring vegetation, rock formations and natural features are not disturbed, modified or removed;
- b) ensuring culturally significant sites are afforded appropriate respect, for example:
 - do not disturb religious artefacts such as alters;
 - do not disturb historical artifacts such as old wells, walls, pottery; etc.,
 - follow instructions on heritage site signage; and
 - obtain the appropriate permission before visiting private or restricted sites;
- c) avoiding introducing or transporting non-native species of flora and fauna (e.g. avoid transporting firewood between areas); and
- d) helping prevent the spread of weeds and pathogens by cleaning, where appropriate, vehicles, equipment, boots and clothing.

B.6 Minimise the impact of campfires

Minimising the impact of campfires can include but is not limited to:

- a) using portable stoves for cooking;
- b) using alternative night-time light sources (e.g. head torches and lanterns);
- c) lighting fires only where and when permitted;
- d) checking with the land owner or land manager for current requirements and restrictions on lighting fires;
- e) minimising the size of the fire to what is needed for cooking or illumination, do not make bonfires;
- f) ensuring firewood is collected from fallen dead wood only where firewood collection is permitted by the land owner or land manager and in accordance with their requirements;
- g) using existing fireplaces where available and avoiding creating new fireplaces; and
- h) dispose of rocks used as fire ring to avoid creating the impression it is an established fireplace.

B.7 Respect wildlife

With only relatively small areas of natural habitat available, wildlife in Singapore have limited amount of the resources they need to survive. Respecting wildlife and recognising that humans are encroaching on their home is important in ensuring the ongoing survival of wildlife. Respecting wildlife includes:

- a) travelling quietly;
- b) educating participants where appropriate about the role a species plays in its environment and the importance of its position within an ecosystem;
- c) not approaching nesting, feeding or breeding sites;
- d) reporting injured wildlife to the land owner, land manager or wildlife authority;

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- e) reporting sightings of known feral animal species and locations of invasive plant species to the land owner or land manager;
- f) ensuring vehicles and equipment are appropriately cleaned to avoid spreading environmental pathogens (e.g. fungal spores) and invasive plant species;
- g) ensuring activities are not conducted in quarantine areas; and
- h) identifying and avoiding areas that are ecological sensitivity or fragile.

B.8 Consideration for others

Showing consideration for other visitors includes:

- a) avoiding impacting sensitive or fragile areas of heritage significance;
- b) applying strategies to avoid overcrowding at popular public areas;
- c) reducing disturbance to other users by minimising the use of:
 - powered generators;
 - amplified sound; and
 - lighting;
- d) managing the noise produced by the activity and participants;
- e) managing activity sites and group congregation areas to reduce disturbance to other users; and
- f) observing the safety of other groups by signalling appropriately to warn of dangers (e.g. snakes, falling rocks).

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Annex C (informative)

First aid kit types

C.1 Group first aid kit

A group first aid kit contains sufficient equipment to address the needs of the activity group, including the activity leadership, participants and anyone else attending the activity. It provides the means to address an injury or medical condition while awaiting advanced medical assistance.

In selecting the type and quantity of medical supplies and medication to go into a first aid kit, the considerations include:

- a) number and age of participants and personnel;
- b) medical conditions of participants and personnel;
- c) nature of activity and location (e.g. water-based, land-based, urban setting, heights);
- d) remoteness of the activity location; and
- e) activity duration.

Tables C.1, C.2, C.3, C.4 and C.5 outline typical items which can be included in a group first aid kit:

Table C.1 – Wound management consumables

S/N	Item	Use
1	Individually wrapped sterile adhesive dressings	Small cuts
2	Bandage crepe (elastic type)	Compression
3	Bandage crepe (light type)	Secure dressings
4	Bandage triangular	Slings and immobilisations
5	Cotton balls	Cleaning of wounds
6	Non-adhesive gauze	Dressing wounds
7	Sterile water or saline	Irrigation
8	Absorbent sterile gauze	Dressing wounds

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Table C.2 – Medical consumables

S/N	Item	Use
1	Alcohol swab	Cleaning of equipment
2	Gloves	Protection against bodily fluids
3	Surgical tape	Securing dressings
4	Paramedic's scissors	Cutting
5	Safety pins	Anchoring of bandage slings
6	Portable splints	Immobilisation of injured limbs
7	Sanitary pad	Managing bleeding/menstruation
8	Sharp pointed tweezers	Removal of splinter
9	Thermometer sheaths	Maintaining hygiene between each usage
10	20 ml syringe	Flushing of wounds
11	Sports tape	Preventing blisters and managing sprain/strains
12	Instant ice pack	Preventing swelling

Table C.3 – Medical non-consumables

S/N	Item	Use
1	Digital thermometer	Taking temperature
2	Emergency blanket	Preventing hypothermia
3	Foldable stretcher	Patient management

Table C.4 – Pharmaceuticals (topical)

S/N	Item	Use
1	Aloe vera gel	Relieving sunburn
2	Burn gel	Relieving burn by open fire
3	Antiseptic cream	Preventing infection
4	Calamine lotion	Relieving itch and rashes
5	Analgesic gel/cream	Relieving muscle strains
6	Anti-itch gel/cream	Relieving itch from insect bites
7	Petroleum jelly	Preventing abrasion and, protecting minor cuts and burns

NOTE – Medication can only be administered by a qualified person, e.g. a registered nurse.

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Table C.5 – Other useful equipment

S/N	Item	Use
1	Waterproof bag	Keeping first aid supplies clean and dry
2	Notebook and pen	Recording of patient vital signs
3	Torch	Providing illumination
4	First aid treatment notes	Using as a reference

C.2 Participant first aid kit

A participant first aid kit is carried by a participant and contains items to address their specific or immediate needs. The typical items which can be in a participant first aid kit include:

- a) personal medications;
- b) strapping tape;
- c) blister dressings;
- d) adhesive medical dressing (small);
- e) sunscreen;
- f) lip balm;
- g) aloe vera gel; and
- h) insect repellent.

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Annex D (informative)

Pathways for recognition of activity leader and assistant activity leader competence in Singapore

D.1 Recognition by Singapore governing bodies such as the OAE Council, outdoor sector bodies and national sports associations

Proof of competence of activity leaders and assistant activity leaders is identified by a governing body associated with the activity through official documentation such as a:

- a) certificate;
- b) accreditation card; or
- c) formal letter of confirmation.

Providers should verify the following:

- a) Qualification is relevant to the roles of activity leader and assistant activity leader;
- b) Qualification is current and remains valid for the duration of the planned OAE activity; and
- c) Applicant's experience is:
 - relevant;
 - recent; and
 - up to date.

Proof of experience can be obtained through the applicant's logbook or similar verifiable documentation.

D.2 Recognition by approved training

Providers accept as proof that its activity leaders and assistant activity leaders have the necessary competence through holding a qualification, from an approved training programme run by an entity which is not covered under D.1.

Providers should verify the following:

- a) Training programme is approved by a governing body of the activity;
- b) Qualification is relevant to the roles of activity leader and assistant activity leader;
- c) Qualification is current and remains valid for the duration of the planned OAE activity; and
- d) Applicant's experience is:
 - relevant;
 - recent; and
 - up to date.

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Proof of experience can be obtained through the applicant's logbook or similar verifiable documentation. Governing bodies should have a robust quality assurance process to assess and recognise the training providers, practitioners or programmes under their purview. This process should include:

- a) assessment of training curriculum referenced to industry best practice and regional standards;
- b) verification that the training provider/practitioner's experience is:
 - relevant;
 - recent; and
 - up to date;
- c) a committee of qualified industry peers to conduct the assessment process;
- d) verification of quality and competence by one or more industry peers;
- e) formal documentation of the assessment process and result; and
- f) regular reviews to maintain the quality of training and competency, failing which the recognition can be withdrawn.

D.3 Recognition by peer verification process

Providers use qualified peers of the activity leaders and assistant activity leaders to verify their competence. For example, prospective activity leaders and assistant activity leaders demonstrate skills while being observed by a peer assessor. If the required level of competence is demonstrated, the peer assessors confirm this in writing.

The process of peer verification should include:

- a) benchmarks based on industry best practice;
- b) verification that the applicant's experience is:
 - relevant;
 - recent; and
 - up to date;
- c) qualified peers to conduct the verification process;
- d) verification of competence by one or more peers involving:
 - a practical demonstration of the required skill(s) by the applicant; and
 - demonstrated knowledge of the related concepts;
- e) formal documentation of the peer assessment and result; and
- f) regular peer assessments to maintain the competency.

Where a recognition through a formally accepted process exists, that process should be used rather than a peer recognition process. However, the peer recognition process can provide a reasonable pathway to determining the competence of activity leaders and assistant activity leaders in situations such as:

- a) where no formally accepted process such as those detailed in D.1 and D.2 exists; or
- b) where a prospective activity leader or assistant activity leader is unable to produce proof of an approved training qualification but can demonstrate considerable experience through their logbooks or other documented proof.

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D.4 Examples of applying the recognition pathways

- Example 1: A client organisation considers recognition by governing bodies as its only acceptable means of recognising activity leaders and assistant activity leaders.
- Example 2: A small provider determines that it may use a leader registration scheme administered by a national sports association as its primary recognition pathway for hiring climbing instructors for its climbing activities.
- Example 3: A provider considers that a training qualification is the basis for employment, and supplements this with peer verification during a candidate's period of probationary employment.
- Example 4: A practitioner seeks employment with a provider using his logbook and training qualification certificate issued by a training provider that is accredited by an overseas body. The provider subjects the practitioner to peer verification for proof of competence before hiring.

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Annex E (normative)

Safety support for improvised rafting in different open water environments

Environment	Minimum level of safety escort	Ratio of safety craft to raft(s)	Rescue plans for common incidents ^a	Licences and safety competencies
1. Open water (coastal) – wind, tide, waves, temperature	For distances beyond 20 m from shore – motorised craft.	1 rescue craft ^b to max. 1 raft; or 1 motorised boat to max. 2 rafts.	Tow raft to safe areas; Pick up and transfer pax out of water quickly.	- Valid boat drivers' licence ^c; - Certification for CPR and the use of an AED ^d; - Standard first aid; - Water recovery and rescue skills.
	For distances less than 20 m from shore – no safety craft required, provided an activity leader is onboard the improvised raft.	1 activity leader onboard 1 raft; or 1 rescue craft ^b to max. 1 raft; or 1 motorised boat to max. 2 rafts.	Maintain < 20 m from shore; Take direct charge to steer the raft; Assist participants to swim/row toward nearest put-out point.	- Proficiency in executing 'TX' and 'kayak to swimmer' rescues; - Certification for CPR and the use of an AED ^d; - Standard first aid; - Water recovery and rescue skills.
2. Waterways and lagoons (no more than Grade 1) – some wind, some tide, obstructions in water	For distances beyond 20 m from designated beach up point – motorised craft.	1 rescue craft ^b to max. 1 raft; or 1 motorised boat to max. 2 rafts.	Push raft out of hazardous areas; Direct raft to middle of waterway channel; Assist participants to swim/row toward nearest put-out point.	- Proficiency in executing 'TX' and 'kayak to swimmer' rescues; - Certification for CPR and the use of an AED ^d; - Standard first aid; - Water recovery and rescue skills.
	For distances less than 20 m from 20 m from designated beach up point – no safety craft required, provided an activity leader is onboard the improvised raft.	1 activity leader onboard 1 raft; or 1 rescue craft ^b to max. 1 raft; or 1 motorised boat to max. 2 rafts.		

^a For example: raft breaking-apart; person overboard; raft drifting to danger; sudden bad weather.

^b Rescue craft include kayaks, canoes, inflatable boats and other non-motorised but stable paddle craft suitable for towing a raft during a rescue in the water.

^c In Singapore, the valid boat drivers' licence is referring to the Powered Pleasure Craft Driving Licence (PPCDL).

^d In Singapore, the SRFAC is the accrediting organisation for CPR and AED certification.

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Environment	Minimum level of safety escort	Ratio of safety craft to raft(s)	Rescue plans for common incidents ^a	Licences and safety competencies
3. Inland lake (quarry, reservoir) - freshwater, placid, deep water; entanglement in water.	Activity leader on-board raft or safety craft.	1 rescuer to max. 1 raft or 1 safety craft to 1 raft.	Maintain < 20 m from shore; Take direct charge to steer the raft; Swim to shore in case of capsize.	• Certification for CPR and the use of an AED ^b; • Standard first aid; • Water recovery and rescue skills.
4. Swimming pool	Activity leader on deck.	1 rescuer to max. 20 participants.	Assist participants to swim/row toward nearest pool edge; Deploy rescue means (pole, buoy, throwline).	• Certification for CPR and the use of an AED ^b; • Standard first aid; • Water recovery and rescue skills.
^a For example: raft breaking-apart; person overboard; raft drifting to danger; sudden bad weather. ^b In Singapore, the SRFAC is the accrediting organisation for CPR and AED certification.				

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- [2] Environment Public Health Act 1987 (Chapter 95, Section 113), Environmental Public Health (Food Hygiene) Regulations
- [3] Personal Data Protection Act 2012
- [4] Road Traffic Act (Chapter 276, Sections 6 and 140), Road Traffic (Motor Vehicles, Construction and use) Rules
- [5] Workplace Safety and Health Act (Chapter 354A)
- [6] Workplace Safety and Health (First-Aid) Regulations

NOTE – The regulations and acts listed above are not exhaustive. Users of the standard need to check with the relevant regulatory bodies on the latest regulatory and statutory requirements.

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NOTE – Hyperlinks are valid at the time of publication.

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SINGAPORE STANDARDS AND TECHNICAL REFERENCES

Singapore Standards (SSs) and Technical References (TRs) are in the form of specifications for materials, products, services and systems, codes of practice, requirements for interoperability, methods of test, management systems, guidelines, nomenclatures, etc.

TRs are pre-SSs developed to address urgent industry demand and are issued for industry trials over a period of time. Comments received during this trial period are considered when a TR is reviewed. TRs can become SSs after the trial period, continue as TRs for further industry trials or be withdrawn.

To ensure adequate viewpoints are considered in the development and review of SSs and TRs, committees and working groups set up by the Standards Council consist of representatives from various key stakeholders which include industry associations, professional bodies, academia, government agencies and companies. SSs are also put up for public comment before publication.

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The content of this Singapore Standard was approved on 3 October 2024 by the Safety and Quality Standards Committee (SQSC) under the purview of the Singapore Standards Council.

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SQSC consists of the following members:

	Name	Representation
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Deputy Chairs	: Mr Ong Liong Chuan	<i>Individual Capacity</i>
	Mr Seet Choh San	<i>Individual Capacity</i>
Advisor	: Mr Chan Yew Kwong	<i>Individual Capacity</i>
Secretary	: Ms Dorothy Er	<i>Singapore Manufacturing Federation – Standards Development Organisation</i>
Members	: Ms Delphine Fong Hung Ying	<i>Sport Singapore</i>
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	Ms Adeline Kooi	<i>National Trades Union Congress</i>
	Mr Bernard Kwok Hon Mun	<i>Ministry of Manpower</i>
	Mr Danny Lien	<i>Singapore Productivity Association</i>
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	Ms Brenda Zhuang Sijia	<i>Tan Tock Seng Hospital Pte Ltd</i>

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SQSC set up the Technical Committee on Personal Safety and Health to oversee the preparation of this standard. The Technical Committee consists of the following members:

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Deputy Chair	: Mr Lim Cheong	<i>Individual Capacity</i>
Secretary	: Ms Phoebe Tay Siew Wei	<i>Singapore Manufacturing Federation – Standards Development Organisation</i>
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	Ms Christina Chua Wee Ling	<i>Singapore Institution of Safety Officers</i>
	Mr Darajit Daud	<i>SGS Testing & Control Services Singapore Pte Ltd</i>
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The Technical Committee set up the Working Group on Outdoor Adventure Education to prepare this standard. The Working Group consists of the following experts who contributed in their *individual capacity*:

	Name
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Secretary	: Ms Phoebe Tay Siew Wei
Members	: Mr Abdul Shukor Haji Ramli Mr Ahmad Lutfi Maula Shamsudin Ms Chew Wee Ling Mr Jacob Ho Wen Yih Mr Low Sung Sheng Mr Lye Yen Kai Mr Graeme Conrad Oei Wei Phing Mr Ong Bok Liong Ms Celine Quek Ming Chwee Mr Oliver Peter Sampson Mr Selamat Jufri Bin Ahmad Mr Syed Ismail Bin Sayadu Sultan Ms Tan Ai Chin Ms Wendy Tan Li Qing

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Members : Mr Tony Tan
Ms Ung Bee Koon
Mr Yang Zhongwen

The organisations in which the experts of the Working Group are involved are:

CAMP CHALLENGE Pte Ltd
Explora Pte Ltd
Forest School Singapore
Innotrek Pte Ltd
Institute of Technical Education
Ministry of Education
National Youth Council
Outdoor Learning and Adventure Education Association
Outward Bound Singapore
People's Association
Pivotal Learning Pte Ltd
Republic Polytechnic
Singapore Polytechnic
Singapore Sport Climbing and Mountaineering Federation
Singapore University of Social Sciences
Sport Singapore
The Singapore Scout Association
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